NAME: Shamir Asgarali

PROGRAMME: Business with Financial Management

STUDENT NUMBER: 19024353

SUPERVISOR: José Luis Ruiz del Portal Tranche

TITLE: The resilience of the European economy during COVID-19, the global financial crisis, and the sovereign debt crisis

DATE: 12-09-2021

Dissertation submitted in partial fulfilment of the requirements of the Masters Degree in

Business with Financial Management

University of Northumbria at Amsterdam



Declarations

I declare the following:

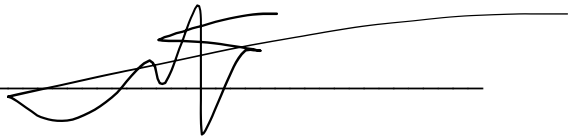
(1) that the material contained in this dissertation is the end result of my own work and that due acknowledgement has been given in the bibliography and references to **ALL** sources be they printed, electronic or personal.

(2) **the Word Count of this Dissertation is 14036 words**

(3) that unless this dissertation has been confirmed as confidential, I agree to an entire electronic copy or sections of the dissertation to being placed on the e-Learning Portal, if deemed appropriate, to allow future students the opportunity to see examples of past dissertations. I understand that if displayed on the e-Learning Portal it would be made available for no longer than five years and those students would be able to print off copies or download. The authorship would remain anonymous.

(4) I agree to my dissertation being submitted to a plagiarism detection service, where it will be stored in a database and compared against work submitted from this or any other School or from other institutions using the service. In the event of the service detecting a high degree of similarity between content within the service this will be reported back to my supervisor and second marker, who may decide to undertake further investigation that may ultimately lead to disciplinary actions, should instances of plagiarism be detected.

(5) I have read the Northumbria/Newcastle Business School Policy Statement on Ethics in Research and Consultancy as detailed in The Newcastle Business School Masters Dissertation Guidelines and I declare that ethical issues have been considered, evaluated and appropriately addressed in this research.

SIGNED*: __Shamir Asgarali_____ 

DATE: ___12-09-2021_____

Abstract

The aim of this dissertation is to investigate Europe's economic resilience based on COVID-19 and past crisis situations, the global financial crisis and sovereign debt crisis. The methodology takes a quantitative comparative approach to investigate how systemic risk behaves during these crisis scenarios, by examining spillover channels on both country, and sectoral level. By identifying these spillover channels, we can discover which channels are contributors, and receivers of shock transmissions so policy makers and investors can make forecasts based on these outcomes. Calculations are made with DCC-GARCH (1,1) model which is a common measure in econometrics for various economic applications. With the help of daily returns of country, and sector data the DCC-GARCH (1,1) model measured that banking, oil and gas, and basic resources are the largest contributors of spillover during COVID-19. Healthcare, food and beverage, and telecommunications are most exposed to spillover during COVID-19. Sectors that are controlling the economy during all crisis scenarios are basic resources and banking. Sectors that are most exposed to spillover transmissions during all crisis scenarios are food and beverage, utilities, and healthcare. In terms of prominent transmitters of spillover on country level, Spain, Italy, and Austria are the largest transmitters of spillover during the GFC and SDC whereas, during COVID-19 prominent transmitters of spillover are Italy, Germany, and Belgium, which makes Italy a prominent common transmitter of all scenarios. Countries most exposed to spillover transmissions during the GFC and SDC are Denmark, Switzerland, and Iceland, while during COVID-19 these countries are Iceland, Switzerland, and Denmark. All crisis scenarios have Iceland and Switzerland as most exposed countries in common.

Table of Content

Declarations 2

Abstract 3

Acknowledgement 4

Glossary 7

1. Introduction 9

1.1 Research motive 9

1.2 Research objectives 12

1.3 Research questions 12

1.4 Content of this dissertation 13

2. Literature review 15

2.1 Intro on literature review 15

2.2 Fundamental principles of systemic risk 15

2.3 Understanding the spillover effect 16

2.4 Relevant measures of systemic risk and spillover effect 17

2.5 The Global Financial Crisis and Sovereign Debt Crisis 19

2.6 Related qualitative studies on systemic risk and the spillover effect during COVID-19 22

2.7 Related empirical studies on systemic risk and the spillover effect during COVID-19 23

2.8 Corporate and government publications on systemic risk during COVID-19 26

2.9 Literature discussion 28

2.10 Literature gaps 30

3. Methodology 32

3.1 Introduction 32

3.2 Research question and hypothesis 32

3.3 Research paradigm 33

3.4 Ethical considerations 33

3.5 Data collection 33

3.6 Methodology 35

3.7 Limitations 37

4. Results and analysis 38

4.1 Introduction 38

4.2 Results and analysis on sectors 38

4.3 Results and analysis on countries 45

5. Conclusions 51

5.1 Introduction 51

5.2 Answer to research questions 51
5.3 Re-evaluation of literature 53
5.4 Limitations and recommendations 55
References 57
Appendices 62
Appendix A: Reflective learning statement 62
Appendix B: Value at Risk 63
Appendix C: Conditional Value at Risk 63
Appendix D: Vector Autoregression 64
Appendix E: DCC-GARCH (1,1) Part 1 65
Appendix E: DCC-GARCH (1,1) Part 2 66
Appendix E: DCC-GARCH (1,1) Part 3 67
Appendix E: DCC-GARCH (1,1) Part 4 68
Appendix E: DCC-GARCH (1,1) Part 5 69
Appendix F: Historical crises 70

Glossary

Abbreviations used:

AEX = Amsterdam Exchange Index

ARCH = Autoregressive Conditional Heteroskedasticity

ATHEX = Athens Stock Exchange

ATX = Austria Traded Index

BEL20 = Belgium Stock Exchange

CAC 40 = Cotation Assistée en Continu, or Paris Stock Exchange

CDOs = Collateralized Debt Obligations

CoVaR = Conditional Value at Risk

DAX = Deutscher Aktienindex, or German Stock Index

DCC-GARCH (1,1) = Dynamic Conditional Correlation - Generalized Autoregressive Condition Heteroskedasticity

ECB = European Central Bank

ESRB = European Systemic Risk Board

ETFs = Exchange Traded Funds

FSB = Financial Stability Board

FTSE100 = Financial Times Stock Exchange 100

GARCH = Generalized Autoregressive Condition Heteroskedasticity

GDP = Gross Domestic Product

GFC = Global financial crisis

IBEX35 = Spanish Exchange Index

IFRS = International Financial Reporting Standards

IMF = International Monetary Fund

ISEQ = Irish Stock Exchange Quotient

LTRO = Longer-Term Refinancing Operations

MBSs = Mortgage-Backed Securities

MSCI = Morgan Stanley Capital International

NPL = Non-Performing Loans

OMXC20 = Copenhagen Stock Exchange

OMXHPI = Helsinki Stock Exchange

OMXI10 = Iceland Stock Exchange

OMXS30 = Stockholm Stock Exchange

PEPP = Pandemic Emergency Purchase Programme

RMBS = Residential Mortgage-Backed Securities

SDC = Sovereign debt crisis

SMI = Swiss Market Index

TCI = Total Connectedness Index

VaR = Value at Risk

VAR = Vector Autoregressions

WHO = World Health Organization

WIG20 = Warsaw Stock Exchange

1. Introduction

1.1 Research motive

Systemic risk occurs when, due to economic changes or catastrophic circumstances, there is a possibility of default within a sector or specific firm that can carry repercussions to other sectors of the economy (Casu, 2015). Spillover effect points out the consequence that unrelated circumstances in one country might carry on specified sectors of other countries. The financial crisis and COVID-19 are two major circumstances in the global economy that impact systemic risk. Since the comparative research on systemic risk and volatility spillover between three catastrophic scenarios are still scarce, this study aims to answer the main question: How resilient is the European economy during crisis periods based on systemic risk, and the volatility spillover between countries and economic sectors.

The global financial crisis and the burst of the housing bubble was ignited in 2001 by the Federal Reserve's monetary policy on reducing the Fed-rate as its objectives were to increase the money supply, decrease borrowing rates, and to enable further investments to stimulate economic growth. As a result, demand for mortgages was rising, which eventually inflated prices on real estate (Valdez & Molyneux, 2015). In 2006, US banks had no restrictions at all on giving out mortgages to sub-prime lenders with no income nor employment status. As the Fed-rate was increasing in 2007, sub-prime lenders were defaulting on their mortgage since they were unable to pay their duties. Until the market saturated, banks were able to sell these houses with a profit. Thereafter, the housing bubble burst, and house prices declined. Eventually, this created a domino effect and selling of real estate occurred on a massive scale (Valdez & Molyneux, 2015). Eventually, the real estate market ignited the financial crisis. However, the level of systemic risk was derived mostly from financial institutions because of the erroneous structure. Systemic risk was caused due to contaminated channels throughout the whole capital market. Complex financial instruments such as subprime mortgages promptly impacted other financial instruments such as Mortgage-Backed Securities (MBSs), Collateralized Debt

Obligations (CDOs), and loans to companies. This showed that the financial system was fragile due to the complexity of financial markets (Crotty, 2009).

The novel coronavirus (COVID-19) made on early January 2020 41 victims and were admitted to Chinese hospitals (Huang, et al., 2020). Although COVID-19 spread swiftly in the region of Wuhan, it used to be neglected by many political leaders. Nevertheless, all political leaders have been warned by intelligence services of a possible catastrophic event (Washington Post, 2020). To control the virus, Wuhan initiated a strict lockdown and the number of infections maintained around 80,000 by mid-February (ECDC, 2020). Unfortunately, travellers around the world had already brought the virus to all regions in the world and COVID-19 reached 146 countries by mid-March 2020 (Anderson et al, 2020). Consequently, the number of infections increased rapidly and, by 15 April, over 2 million infected were reported over 200 countries (ECDC 2020). To prevent the virus from spreading, nations initiated hefty measures such as lockdowns, social distancing, closure of educational institutions, obligatory work from home, cancelling events (e.g., concerts, sport events and festivals) and many more. Moreover, travel restrictions, regional and international, instantly afflicted many sectors of the global economy (FCO, 2020). From February 2020, investors were positive about a quick end of the outbreak. The number of infections kept rising and optimism began to decline. The World Health Organization (WHO hereinafter) announced COVID-19 a pandemic and the global financial market embraced itself for substantial losses. As fear spread among the global economy, investors and traders were deterred from buying stocks while selling stocks occurred massively (Statista Research Department, 2021)

In Asia the fear of the pandemic caused a downward market trend. From the rise of the pandemic, the Shanghai Stock Exchange Composite Index, and the Nikkei Stock Average Index both plummet. Japan had to suffer a tougher blow as the Tokyo 2020 Olympic Games had to be postponed. The fear also struck the U.S economy as the S&P 500 Index, the Nasdaq Composite Index and the Dow Jones Industrial Average Index had the most immense gains and losses in history. A phenomenon that indicates how volatile financial markets can be. All these unforeseen circumstances in the US have led the Senate to develop a two trillion U.S dollar COVID-19 aid bill to strengthen the US economy. In the UK the Financial Times Stock Exchange 100

Index (FTSE100) endured its steepest decline since 1987, and stock exchanges across Latin American suffered the same unfortunate fate (Statista Research Department, 2021).

It is obvious the COVID-19 pandemic and the 2008 financial crisis are strong anomalies in the financial market. The aftermath of the pandemic is expected to take longer than two years as was the case after the financial crisis. Meanwhile investors remain vigilant how the situation develops before investing in stocks (Statista Research Department, 2021). Investors and portfolio managers are facing turbulent times as securities become more volatile. Hence, new investment strategies must be adopted to mitigate risk as much as possible. For this reason, the unforeseen pandemic has brough attention for a deeper analysis of co-movements between sector- and the accumulative of country's stock indices within Europe to assess the interconnection and contagion in extension to recognize underlying ramifications for financial market stability.

The COVID-19 pandemic and the financial crisis are legitimate scenarios that call for further research on systemic risk across different sectors of the European economy. Some research has already been conducted on this topic. Hanif et al (2021) investigated the spillover effect between ten Chinese and U.S stock sectors during the COVID-19 pandemic. This study concludes that pre COVID-19 the U.S creates more spillover than China, but during COVID-19 it turned the other way around and even intensified between March and April 2020. Another study conducted by Amar et al (2020) examined the extent of the connectedness among the MSCI Europe (EUR), MSCI Emerging Markets Latin America Index (LAM), MSCI Emerging Markets Asia Index (EMS), MSCI North America Index (NAM), MSCI GCC Countries Index (GCC) and MSCI Pacific Index (PAS) over a time period between June 2012 and, to 11 May 2020. Amar's (2020) research confirms a disconnect between the LAM and the global financial market, whereas the other mentioned markets act like the downturn of the economy caused by COVID-19. Furthermore, Ferreira et al (2020) were investigating the volatility spillover Among European financial markets and concluded that Sweden, Germany, and the Netherlands are contributing to the highest volatility spillover, whereas the lowest spillover were observed in the Polish and Irish stock market. These discoveries reveal substantial information on the

European stock market interdependence during the pandemic. A study considerably like Ferreira's et al (2020) was investigating the interdependence of the global- and country stock market during COVID-19. Here, Abuzayed et al (2021) uncovered the tail risk spillover among global markets and its main finding shows a substantial degree of integration in the downside risk of the stock market during the pandemic. Abuzayed's et al (2021) data consists of stock market indices of 14 most affected countries. During the pandemic, stock markets in the U.S and Europe spread and obtain a high marginal risk to- and from the global stock market index in comparison with the Asian stock market. Lastly, a study from Laborda & Olmo (2021) made a comparison between COVID-19 and the financial crisis on the volatility spillover between economic sectors within the US market. This study concludes that the banking and insurance sector were the most relevant during the financial crisis, and the energy and technology sector most relevant during the pandemic.

This dissertation will investigate how system risk develops across Europe on a country- and sectoral level, and on financial institutions, by comparing it with three crisis scenarios: the global financial crisis (GFC hereinafter), sovereign debt crisis (SDC hereinafter) and COVID-19. Hence, this study will aim to identify spillover transmission channels and identifying how current economic foundations can prevent risk transmissions during the COVID-19 pandemic.

1.2 Research objectives

To investigate spillover transmissions channels, this research is built on three main objectives namely, to discover prominent spillover channels of both countries, and sectors during COVID-19. The second objective is to discover the most notable receivers of spillover of both countries, and sectors COVID-19. The third objective is to find commonalities of transmission channels between all crisis scenarios.

1.3 Research questions

To answer this dissertation's primary question "how resilient is the European economy during crisis periods based on systemic risk, and the volatility spillover between countries and economic sectors?", the following three sub-questions are presented below:

Research question one: What European countries and sectors contribute most to spillover transmission during COVID-19
Research question two: What European countries and sectors are most exposed to spillover transmission during COVID-19
Research question three: in terms of volatility spillover, are there commonalities between the GFC, SDC and COVID-19 of volatility transmission channels on a country, and sectoral level?

1.4 Content of this dissertation

This dissertation is divided in the following sections:

Section one dives into the motive behind this dissertation's topic while presenting important background information on systemic risk, and why it is an enormous concern for the European economy as the break-out of the COVID-19 pandemic caused a tremendous impact. Some exemplary studies based on Europe, and other parts of the world, are presented to demonstrate what impact the COVID-19 pandemic carries among different spectrums. This section raises three objectives, three sub-questions based on these objectives, that ultimately should answer this dissertation's main questions.

Section two serves as an in-depth literature review to bring more perspective on systemic risk, volatility spillover and measures researchers and financial analysts use to detect spillover transmissions channels. A comprehensive understanding on these aspects is prerequisite since this section describes empirical, and qualitative studies, but also articles provided by recognizable financial, and governmental institutions. Also, noteworthy is that most empirical studies implement similar, or same methodologies since outcomes from these studies will be supplementary for this dissertation's conclusion. Eventually, based on all described literature, gaps on existing will be highlighted. Ultimately, based on these gaps, section two describes how this dissertation will add to the existing literature.

Section three describes the hypothesis based on section two's research questions. These hypotheses are based on findings from previous studies. Next, an explanation

is given on Northumbria University's green status of the ethical approval application. The university's approval on data collection is mandatory protocol before initiating data collection and research. Next, this section dives into rationales for the selected data collection method. Also, an explanation about positivism as the appropriate research paradigm is explained. Positivism often relates to a quantitative approach of which the methodology of this dissertation is based on. Lastly, limitations regarding this dissertation are addressed.

Section four presents an analysis of empirical results that are based on section three's methodology.

Section five congregates arguments from literature described in section two, to combine empirical results from section four. This implies that all related theories are revisited, compared, and contrasted based on outcomes from this research. Consequently, an answer to this topic's main question will be given based on all three sub-questions. Next, outlines on recommendations and suggestions for future studies will be presented. Finally, potential limitations will be highlighted.

2. Literature review

2.1 Intro on literature review

This section will discuss key literature relevant to the objectives of this dissertation. Initially, some key principles on systemic risk (section 2.2) will be explained along with an understanding on volatility spillover (section 2.3). In addition to systemic risk and the spillover effect, a series of imperative formulas (section 2.4) to quantify these elements will be illustrated. Moreover, a course of key literature on how systemic risk behaves during the GFC (section 2.5), SDC (section 2.6) and COVID-19 (section 2.7) will be presented, followed by a selection of key publications disclosed by financial- and governmental institutions (section 2.8). At the end of this section, a discussion on all literature (section 2.9), and literature gaps (2.10) will be presented.

2.2 Fundamental principles of systemic risk

Systemic risk is a fundamental term in finance and economics and there are several definitions on what systemic risk implies. One definition from Bernanke (2012) explains systemic risks derived from economic changes that can potentially jeopardize the financial system and eventually the whole economy. This phenomenon does not emerge solely from one or two institutions. According to ex-president Trichet (2009) of the European Central Bank (ECB from hereafter), developments that cause systemic risk may consist of unexpected events of failing associated financial institutions causing endogenous inequality. Fouque and Langsam (2013) state that systemic risk emerges from disturbance in the economy which decrease capital flows and eventually impact on negative GDP growth on a global scale. Reinhart et al (2009) agrees with Fouque, Langsam (2013) and Trichet (2009) on the downward effect of the composite GDP index. In addition, Reinhart gives more perspective on the contagion spreading from one or more financial centres to other countries. The Financial Stability Board (FSB hereinafter, 2011) gives more perspective on the faulty financial system that cause turmoil regarding financial services. Consequently, failures in the financial system can potentially cause substantial damage to the entire economy. The bottom line according to the

FSB (2021), is the sentiment of unfavourable externalities deriving from faulty investment vehicles, institutions, or markets. Acharya et al (2016) explain that having insufficient capital to cover its liabilities regarding the financial sector, has the potential to cause systemic risk. Eventually, it can contaminate the capital market and other financial institutions. At the same time, deterioration occurs on financial intermediation, both in lending and borrowing whether it regards to corporations or households (Acharya et al, 2016). The International Monetary Fund (IMF hereinafter, 2018) depicts that systemic risk comes from a mixture of credit risk, market risk, liquidity risk and operational risk. Additionally, a notion very similar on risk contagion in alignment with Acharya et al (2016) and Reinhard et al (2019), the IMF brings perspective on idiosyncratic shocks from financial institutions that eventually contaminate other organisations in the financial sector.

As we can conclude from various theories that characterize systemic risk, it is a common underlying risk that carries the potential to substantially damage the financial constructure and the entire economy. On the other hand, there are circumstances on which the impact is limited to a certain sector or geographic region but does not spill over other regions or sectors (Gottesman & Leibrock, 2017). However, a key aspect of this research is called the spillover effect which depicts the network effect between economies across the world, caused by a catastrophic event which can be a financial crisis or a pandemic such as COVID-19. This will be further explained in section 2.3

2.3 Understanding the spillover effect

For every professional in the financial sector, the attention on monitoring and controlling financial risk increased significantly over the past decades. When assessing financial risk, possibilities on detrimental market fluctuations have always been a great concern. Market fluctuations take place on certain occasions such as spillover from other countries or sectors, regulatory reforms, surprise shocks or natural disasters. As one of these calamities occur, a chain-effect follows while market fluctuations can deplete large amounts of investors capital, followed by bankruptcy of financial institutions, which eventually may cause a breakdown in the financial system and simultaneously lead to social unrest. Due to all the devastating

experiences that occurred from adverse market fluctuations, investors are now very vigilant on large market fluctuations (Liu, Liu, Hong, & Wang, 2015). An example dating from October 19, 1987, on Black Monday, the US stock markets dropped down by 23 percent, causing a downturn of one trillion USD worth of capital (Bernhardt & Eckblad, 2013). Another example between 1989 to 1992 comes from the Japanese stock market when the Nikkei 225 index declined from 39.000 to 17.000 Yen which resulted in a capital loss off 2.7 trillion USD (Shiller et al, 1996). Final example, the debacle during 1994, the US Federal Reserve Board initiated a sequence of six interest rates hikes in perpetuation that eliminated 1.5 trillion USD in global capital. All which occurred after keeping interest rates low for a duration of three years. Consequently, a chain reaction occurred, starting from the US Orange County Investment Pool, which held a 7.5 billion USD portfolio of municipal investors (cities, counties, schools), and lost over 1.64 billion USD in December 1994.

2.4 Relevant measures of systemic risk and spillover effect

The first model, related to this topic, is value-at-risk (VaR Hereinafter). VaR is a widely accepted method in risk management to quantify market risk. It estimates a supply of money, and it measures probabilities. Therefore, it is rather considered a model to estimate a level of confidence prior given by the user. VaR calculates an assets volatility, meaning that higher volatility equals a higher chance of loss. VaR is a method to analyse a bank's risk vulnerability, by giving a volatility estimate, so that risk managers can have an estimation. Calculating VaR consist of four steps: First, establish a time horizon the company considers a possible loss. Second, select a level of certainty, also known as confidence level, that is appropriate for the VaR calculation. Third, construct a probability distribution of possible returns for the portfolio, or asset subject to evaluation. Fourth, calculate the VaR appraisal by analysing the loss amount below the confidence interval (Choudhry, 2006). Importantly, three factors should be considered for VaR: Minimum loss, a specific period, and the possibility of surpassing a given loss (Akshay, 2018). Appendix B shows a definition of VaR.

As we are now familiar with VaR, understanding conditional Value at Risk (CoVaR hereinafter) is one prerequisite along with its purpose as a measure of systemic risk.

CoVaR essentially estimates how financial institutions are adding to systemic risk, plus its participation on spillover to other financial institutions. CoVaR, along with other systemic risk methodologies, can assist in gaining a finer understanding of any risks jeopardizing the financial system. CoVaR, or expected shortfall, is designed to forward the limitations of VaR, namely accounting only a certain percentile of losses and does not look or provide any insight at all at what losses occur if the situations turn worse than the assumed threshold. Hence, VaR only calculates the cut-off for example 5 percent scenarios, and would not give any insight at all if losses go further than the left-hand of the distribution. CoVaR will tell what the average loss is and what is expected to be endure, for any given percent, of worst-case outcomes. Additional reasons for preferring CoVaR are as follows: CoVaR takes system-wide risk levels into account, in particular spillover effect among financial institutions and determine to what extent the institution adds to systemic risk. Also, CoVaR estimates the volatility to the financial system when institutions face adversities, and in a regular state. CoVaR also measures risk exposure for institutions during crisis periods. Lastly, CoVaR can be used to estimate changes in systemic risk based on financial market data (Adrian & Brunnermeier, 2016). Calculating CoVaR becomes more straightforward after VaR has been calculated. Appendix C illustrates the definition of CoVaR.

The Autoregressive Conditional Heteroskedasticity (ARCH Hereinafter) model is a statistical framework that anticipates future volatility by evaluating time series volatility. In financial risk, ARCH is applied to anticipate risk by arranging a volatility framework that simulates actual markets. ARCH exhibits that times of high volatility are pursued by increased high volatility and times of low volatility are pursued by increased low volatility. As a result, volatility is inclined to cluster, which is beneficial for risk managers when contemplating the risk of certain assets over different periods. The conditional heteroskedasticity component of the ARCH model points out the evidence of the nonconstant volatility in financial markets. Meaning that financial data such as stocks, GDP, or foreign exchange rates, constantly face stages of low and high volatility. ARCH arranges a framework that risk managers can utilize, rather than assuming an average or constant for volatility. Furthermore, ARCH models are best applied with high-frequency data such as hourly, weekly, or monthly data, which makes the model suitable for financial data. Hence, ARCH

models have become one of the most universal methodologies for quantifying financial market volatility (Kenton, 2020).

The Generalized Autoregressive Condition Heteroskedasticity (GARCH hereinafter) is an amplification of the ARCH model that embeds a moving average segment along with the autoregressive segment. To be specific, the GARCH model incorporates the considerations if it should create the white noise residual of an additional measure, also called lag variance terms, plus the incorporation with lag residual errors. Due to the moving average segment GARCH is able to simulate the conditional difference in variance, as well as differences in time-dependent deviation (Brownlee, 2019).

The Vector Autoregression (VAR hereinafter) is an algorithm used to forecast, when multiple timeseries impact each other. This implies that the connection of two time series is bi-directional, an ability which previously mentioned models do not possess. There are two pre-requisites to use VAR: First, two time series are needed. Second, both time series must be correlated. The term autoregressive illustrates that each time series are functions of past values, meaning that predictions stand for delayed time values of the series (Prabhakaran, 2019). An example on using VAR is to analyze how real GDP influences policy rate, and likewise, how policy rate influences real GDP (Erica, 2021). Appendix D illustrates the definition of VAR.

2.5 The Global Financial Crisis and Sovereign Debt Crisis

Since money and financial markets came to existence, financial crises have always occurred.
However, every financial crisis over the past hundred days shows different symptoms than the crisis before. As the financial industry and its services continuously change in terms of market structure, risk analytical tools, technological breakthroughs and regulatory frameworks, the financial industry of today shows little resemblance to the one 50 years ago (Gottesman & Leibrock, 2017). A timeline containing all historical crises from 1636 to 2008 can be found in appendix F. This dissertation will mainly focus on the GFC, SDC and COVID-19.

During the financial crisis of 2007, the breakdown of many financial institutions worldwide generated repercussions across the world economy and financial markets. This led to a recession over Europe and the U.S. A couple of years before the financial crisis, some high-profile public figures countered, or ignored the concern about a collapse in the U.S financial system. Between 1970 and 2007, the US financial sector doubled from four to almost eight percent of Gross Domestic Products (GDP hereinafter, Gottesman & Leibrock, 2017). Also, since 2000, financial innovation was rapidly developing while Wall Street was profiting heavily on new types of securitization products. From 2003, US house prices increased by 27 percent and eventually Residential Mortgage-Backed Securities (RMBS hereinafter) were generated with a total worth of 4 trillion USD from which Wall Street generated mortgage-backed CDOs worth 700 billion USD.

The blowout of the real estate bubble, bankruptcy of companies and price collapse induced an enormous shock to the economy. Consequently, a vast number of banks acquired excessive amounts of bad debt in their financial statements. The disastrous consequences of the GFC created a world-wide recession, impacting private and public sectors, as well as wages and asset prices, as was affirmed by European countries (Glod, 2018). Accordingly, the GFC impaired the European economy while jointly raised debt levels of many EU member states (Glod, 2018). Simultaneously, European banks were afflicted by the GFC.

One factor why European banks were vulnerable to contagion of the GFC was primarily due to globalization in the banking sector from Europe to the US market (Welfens, 2008).

The Eurozone debt crisis emerged from several causes. First, the absence of a competent structure and institutions in concern to impede the stack-up of macro-economic inequalities. Second, the absence of European institutions which should involve in assimilating shocks. As the cost of borrowing was lowered, followed by the access in the Euro-zone, a large capital inflow to the intra-eurozone was caused (Wijffelaars & Loman, 2015).

De Bruyckere et al (2013) investigated how contagion between sovereign default risk and banks appear between 2007 and 2012. This study states that banks with fragile funding structure, less conservative banking activities, and weak capital buffers are

notably prone to risk spillover. Furthermore, when looking at the country level, to most substantial factor of contagion is the debt ratio. Lastly, as De Bruyckere et al (2013) conclude, the most effective measure the government can enforce is capital injections to minimize spillover intensity.

A study by Santis (2013) concludes that the spillover effect from Greece has played an important role on creating contagion to countries with fragile fiscal policies, higher dependency of foreign financing and an inadequate level of competitiveness. To elaborate, at the end of 2009 when Greece's fiscal problems were disclosed, the debt crisis spread to most EU member states which eventually, created a challenging assignment for the European Monetary Union. Causes of the sovereign debt crisis are related to country-specific risk, contagion risk, spillover risk, and aggregate risk (Santis, 2013). Aggregate risk is caused by developments in monetary policy, risk aversion and global unpredictability. Country-specific risk is associated to default risk on sovereign debt, the capability to boost funds in the primary market and liquidity aspects in the secondary market (Santis, 2013). Keeping all mentioned risks separated allows for better perspective on creating new policies since central banks can intervene more successfully in case the economy becomes subject to risk contagion, or liquidity problems. The sovereign debt crisis brought three topics to debate, namely flight to liquidity, the transmission of spillover effect and sovereign solvency risk.

Santis (2013) points out two key factors that contributed to the European sovereign yield spread namely: The spillover effect from Greece and flight to liquidity favouring the Bund.

Flight to liquidity, which was developed with the spread between the Bund and a German state agency, is the only particular reason of the spread between the Netherlands and Finland.

The spillover effect that originates from Greece, mostly caused by rating downgrades and foreign financial support, played a key role on developing sovereign spreads. Also, Santis' (2013) research disclosed a "wake-up call" on risk formation as the GFC exposed countries with fragile economic foundations. These countries contributing to further spillover during the SDC are Spain, Belgium, Portugal, Austria, Ireland, France, and Italy.

2.6 Related qualitative studies on systemic risk and the spillover effect during COVID-19

This section will discuss the most related qualitative studies closely related to COVID-19, systemic risk and the spillover effect. An article from Goodell (2020) brings justification of researching the pandemic and its devastating economic impact to light. Goodell's (2020) anticipatory article also narrates that current research on COVID-19 and its economic impact are practically parallel to similar crises events, alongside some differences, should draw more attention for further investigation. Furthermore, Goodell has no doubt that natural disasters can bring the economy to a vulnerable state on a considerable scale and policymakers, participants in the financial sector and the public must be aware of these circumstances. Lastly, Goodell (2020) raises question on how the pandemic will impact financial institutions such as insurance companies, pensions funds, banks, and government- and political stabilization measures. Also, some supportive studies on Goodell's (2020) statement are focussed on the banking industry. Lagoarde-Segot and Leoni (2013) explain that banks in developing countries are vulnerable during an economic downturn on which nonperforming loans (NPL hereinafter) are the main cause. In developing countries, large increases in deposit turnover is often related to the spread of HIV since large scale withdrawals of deposits are often needed for individual treatment (Lagoarde-Segot & Leoni, 2013). Lagoarde-Segot and Leoni (2013) created a theoretical model that presents the increased possibility of a breakdown of the banking industry as the presence of a pandemics rises. Skoufias (2003) addresses the issue on banks in developing countries and microfinancing how these institutions are pressured during times of epidemics as the aggregate shock will impact all lenders.

In times of floods or crop failures, financial institutions in rural areas are often prone to bank runs (Binswanger & Rosenzweig, 1986). A more recent study that scopes on NPL reveals the lessons we have learned based on past events. This study conducted by Ari et al. (2020) is based on 88 banking crises from 1990 in Europe. The data of this research reveals that managing NPLs is crucial to stabilize the economy. Some measures we have learned from the GFC of managing NPLs are provisions on higher bank capital and IFRS 9 accounting standards. However, these

measures work effectively on credit booms, which COVID-19 is not. Therefore, COVID-19 still brings challenges on managing NPLs since public debt rises in the Euro-zone, European banks generate less profit, and finally, a factor that obscures NPL measures, economic sectors that are underperforming. Ari et al. (2020) conclude their research by stating the importance of implementing reformed post-COVID-19 NPL measures. D'Erman et al. (2020) state that the EU must commit to a total collaboration of all the member states as the current pandemic will bring the EU in a new economic crisis. The pandemic will force the EU into new developments, namely a reformed banking union, as well as creating and providing addition goods for the public regarding the Economic and Monetary Union (EMU) (D'Erman et al, 2020).

2.7 Related empirical studies on systemic risk and the spillover effect during COVID-19

The most influential empirical studies already conducted related to this dissertation are focussing on specific countries, financial markets, and economic sectors. It is essential to recognize the current findings of this topic on how COVID-19 influences systemic risk on different fields. A study managed by Shahzad et al. (2021) was examining how the volatility spillover on Chinese stock market sectors presents itself during COVID-19. The sample period is from January 2, 2019, till September 30, 2020. Finding of this study indicates a significant strength of asymmetric spillover among Chinese stock sector indices during the estimated time period. Four sectors need special attention by investors and policymakers namely, industrials, energies, materials, and utilities as these sectors contribute to negative volatility spillover and can potentially destabilize the market. Hence, Shahzad et al. (2021) conclude their research by stating that disastrous events like COVID-19 have a severe impact on volatility spillover among economic sectors in the Chinese market. Another study focussed on the Chinese market conducted by Wang and Li (2021) explains the spillover relation between the crude oil market, the gold market, stock market and bond market by analysing daily observations during 2003 and 2019. The study concluded the following: Of all the markets, the crude oil market, characterized as risky with high returns, contributes to most of the volatility spillover on the long-term. Also, the crude oil market mainly brings long-term volatility spillover on the Chinese

financial market. There is more long-term volatility spillover than short-term. Additionally, positive volatility spillover overrules negative volatility spillover since the crude oil market transfers the largest amount negative volatility spillover. During times of the GFC, the sovereign debt crisis, and Brexit, Wang and Li (2021) showed that values of risk spillover inclined drastically. Since bad volatility is usually related to crisis situations, examining a long-term volatility spillover curve can anticipate signs of risks. Policymakers and investors should pay close attention to crude oil volatility when implementing hedging strategies during times of crises (Wang & Li, 2021).

A study outside the Chinese market conducted by Laborda and Olmo (2021) directs on economic sectors in the U.S by determining volatility spillover and network connectivity measures. Data of this study consist of daily sectoral returns of the S&P500 covered over a period between 2003 and 2020. Among all the sectors that were examined, energy, biotechnology, technology, and banking & insurance are the prime sectors when shocks spread to other economic sectors in the U.S. During the GFC the banking & insurance sector are the most relevant while the energy sector is more relevant during COVID-19 (Laborda & Olmo, 2021). A study by Aslam et al. (2021) sets focus on stock market volatility of twelve European countries. Data consist of observations between December 2019 and May 2020. Findings of this study narrates that Sweden and the Netherlands contribute to the most directional volatility spillover while the least originates from Ireland and Poland. Countries subject to spillover are Belgium and Germany. A study conducted by Ortmans & Tripier (2021) shed light on how the ECB's monetary policy mitigated further spread caused by the pandemic to the European sovereign debt market. One of Ortmans & Tripier's conclusion narrates the counterfactual analysis explaining that spreads would have deluged around March 18, 2020, with 4.2 percent in France, 12.5 percent in Spain, and 19.5 percent in Italy, if the ECB had not engaged their plans stated on the March 12, 2020, press conference by Christine Lagarde.

On this date, the ECB issued a series of monetary policy resolutions to prevent further economic damage caused by the pandemic. Furthermore, after the stock market crash on March 12, 2020, the stock market reached a plateau, but interests rates kept climbing to over 2.8 percent on March 17 (Ortman & Tripier, 2021). The

ECB started a Longer-Term Refinancing Operation (LTRO) to ensure liquidity and issued the commencement of an intervention proceeding called, the Pandemic Emergency Purchase Programme (PEPP), which created a reboot in stock prices and a reversal in sovereign rates. The pandemic may spread contagion from person-to-person, its spread to financial markets ended in Italy and the rest of Europe (Ortman & Tripier, 2021). Ortman & Tripier (2021) support the view that interventions of the ECB in response to COVID-19 have prevented a devastating escalation of sovereign default risk comparable with the sovereign debt crisis. According to the theoretical framework of Arellano et al. (2020) there is a relationship between a rising probability of default in sovereign debt and the COVID-19 pandemic. Ortman & Tripier enforce this statement with empirical findings and suggest that fiscal transfers committed by governments to stimulate consumption, are inhibited due to borrowing capacity and default risk, which eventually, create higher costs to sustain lockdowns. Therefore, according to the framework of Arellano et al. (2020), the higher the pandemics intensity, the higher the probability will be of default on sovereign debt. Also, the ECB illustrates that the pandemic induces an urgent rise in direct costs, primarily to set focus on repercussions facing the public health. However, from a macroeconomic point of view, most of the pandemic's impact relates to lockdown measures, that severely impacts firms, households, workforce, and fiscal measure packages (Ortman & Tripier, 2021). Consequently, the projected governmental budget deficit of the Euro-zone raised from 0.6 percent of GDP to 8 percent of GDP. Risk spillover to the banking sector due to substandard balance sheets was addressed by Schularick and Steffen (2020) and state that banks from certain countries raised their national sovereign debt, which eventually creates anxiety of a re-occurrence of the SDC.

2.8 Corporate and government publications on systemic risk during COVID-19

According to the latest report of the Deutsche Bank (2020), large banks were heavily afflicted by the pandemic. Revenues declined by five percent. The major reason of this impact occurred from a spike in loan loss provisions. Consequently, profits almost dispersed despite banks were cutting costs (Schildbach, 2020). As reported by Gobbo (2020), author and financial risk manager at KPMG, the banking sector is impaired heavily and should prioritize on four areas if risk managers want to support their organisation, but also strive to sustain their operations post COVID-19. First, credit risk can differ for each sector. Airlines, oil, tourism, healthcare, gas, and automotive industries are seen as the most impaired sectors as their operational models tend to have many fractures. Therefore, it is crucial for banks to recognize these sectors and invent solutions like accelerated credit review processes for rapid liquidity. Additionally, on the long-term, banks should integrate a strategy on managing rating models appraisals, while simultaneously examine credit portfolios incorporated with pandemic scenarios. Under these circumstances, banks should watch out for relaxation measures as these opportunities provide for stability. Second, banks should concentrate on strong management of liquidity since it has shown to be substantial to aid in comparable economic drawbacks and developing customer demands. On top of that, liquidity stress models created after the GFC do not offer an ample understanding on present developments on liquidity, so banks are necessitated to re-evaluate certain liquidity frameworks. Third, since the beginning of COVID-19, there has been an increase in market volatility among all types of assets, including market risk capital requirements. Therefore, banks should concern themselves about daily market risk operations, while simultaneously they should look after capital requirement ramifications and various supervisory procedures. The pressured market circumstances may induce financial losses from adjustments on credit valuations. Additionally, a rise in government's prudential provisions can derange bank's capital ratios. Fourth, Gobbo (2020) points out capital management and enterprise risk on how bank's capital adequacy and capital ratios are harmed due to COVID-19. At the same time, weakened credit conditions could ignite a consequential increase in credit deteriorating. Nonetheless, banks are suggested to remain supportive to the economy by extending lending services, which eventually generates higher negative exposure on credit cycle. Gobbo's (2020) suggestion to

banks during COVID-19 is to ensure that their management tools are approved for their purpose. Also, banks should identify a feasible strategy to sustain the crisis. One possibility is to pull-back from their risk appetite provisions and recovery thresholds to encourage the plan of returning back-to-normal.

The European Systemic Risk Board (ESRB hereinafter) provides macroprudential supervision of the European financial system and pursues measures to impede or mitigate systemic risk. The ESRB has a broad scope that includes insurance companies, banks, shadow banks, asset managers, financial markets among others. Whenever the ESRB engages its macroprudential authorization, it controls for signs of systemic risk over every financial institution of each member state, and issues guidance and recommendations. The latest ESRB annual report covers the period 1 April 2020 and 31 March 2021,a period defined by COVID-19. Lagarde (2021), president of the ECB and Chair of the ESRB, states that the European financial systems showed more resilience than the previous GFC, except for the calamities that can build up deficiencies in the financial system. These deficiencies contribute to the impaired financial system when the European economy was reliant on strong liquidity. Between April and May 2020, the ESRB initiated extra measures to reinforce the financial stability by mitigating risk caused by the pandemic (Lagarde, 2021). These measures prioritize on (i) fiscal measures on loan guarantee schemes to preserve the economy; (ii) the impact of market illiquidity for insurers and asset managers; (iii) the influence of downgrading procyclical bonds on financial markets; (iv) the prohibition on dividend pay-outs and share buybacks; and (v) liquidity risk rising from margin calls (Lagarde, 2021). Furthermore, the economic impact of the pandemic heavily affects the banking sector as credit risk is on the rise. The ESRB strives to provide capital buffers, so banks are still able to lend in the damaged economy. Simultaneously, it is crucial to build capacity for both the public and private sector to handle a potential uprise in corporate insolvencies. The latest ESRB annual report points out the main risks identified: (i) a possible soar in insolvencies in the private sector which emerged from the latest GFC; (ii) financial institutions facing macroeconomic threats and (iii) the rapid developing market illiquidity and repricing of risk.

2.9 Literature discussion

About systemic risk on the sectoral level, Shahzad et al. (2021) discovered that during the initial phase of COVID-19, four Chinese economic sectors: Energies, industrials, materials, and utilities are contributors to negative spillover. Strong receivers of spillover are healthcare and industrial goods and services. This affirms that a pandemic creates volatility spillover on certain economic sectors. Wang and Li (2021) conducted a study covering the period of the GFC and SDC on the Chinese market and made several conclusions. Firstly, the crude oil market creates the most volatility spillover on the long-term. Secondly, when the crude oil market gets affected, it immediately creates spillover to the Chinese financial market. Thirdly, negative spillover increases drastically in times of crises and negative spillover should be monitored closely as these are essential indicators of risk. Especially the crude oil sector, policymakers and investors must pay close attention. In the U.S market, Laborda and Olmo (2021) covered all periods of the GFC, SDC and the initial phase of COVID-19, and uncovered that, during times of crises, energy, technology, and banking & insurance are the prominent sectors when shocks escalate to other sectors of the U.S economy. During COVID-19, investors and policymakers should pay attention to the energy sector, while during the GFC and SDC, it was the banking & insurance sector. By comparing how systemic risk behaves during COVID-19, the GFC and the SDC on a sectoral level, we can conclude that the energy and oil sectors, in the U.S and Chinese market, are the largest contributors to negative spillover to other sectors in the economy, whereas during the GFC and SDC it was the financial sector.

Elaborating systemic risk on financial institutions: During the GFC, financial institutions created a large negative spillover across the globe, and eventually created a recession over Europe and the U.S. Before the entire collapse, signs of deficiencies of the financial structure were ignored (Gottesman & Leibrock, 2017). Also, the collapse of the real estate bubble created negative spillover to all other areas in the economy, and banks incurred massive amounts of bad debts (Glod, 2018). Welfens (2008) state that European banks were prone to spillover due to globalization. Whereas De Bruyckere et al (2013) and Gottesman & Leibrock (2017) mention fragile funding structure, weak capital buffers, and less conservative

banking activities, as the main reasons why banks became subject to risk spillover. De Bruyckere et al. (2013) preferred solution to mitigate spillover intensity was implementing government capital injections. On the other hand, Lagoarde-Segot & Leoni (2013) give more perspective on banks in developing countries by declaring NPLs as main contributors of an economic downturn. The study based on 88 banking crises from 1990 conducted by Ari et al. (2020) is consistent with Lagoarde-Segot's & Leoni's (2013) statement on NPLs, elaborating further how provisions on bank capital and IFRS 9 standards work effectively on credit booms, but serves much less for a pandemic like COVID-19. Ari et al. (2020) suggestion post COVID-19 is to implement adjusted NPL measures as the public debt in the Euro-zone keeps rising. A similar notion of Schildbach (2020) brings perspective on spikes in loan loss provisions from large banks. Four provisions to the banking sector suggested by Gobbo (2020) is to (i) implement accelerated credit review processes concerning all inflicted sectors, to stimulate rapid liquidity. Next, (ii) banks should re-evaluate liquidity frameworks since liquidity provisions based on the GFC, appear to be inappropriate for COVID-19. Furthermore, (iii) banks should focus on market risk operations and capital requirement complications due to higher market volatility on many types of assets. Lastly, (iv) as Gobbo (2020) suggests to banks, is to maintain lending services to keep the economy stable, while simultaneously refrain from risky operations. Likewise, Lagarde's (2021) actions on improving liquidity, implementing fiscal measures on loan guarantee schemes and capital requirements, are in alignment with Gobbo's (2020) recommendations.
To conclude how systemic risk differs when evaluating financial institutions, between COVID-19, the GFC and SDC: Globalization, the fragile financial structure, incurring bad debts due to the real estate collapse and weak capital buffers are main contributors during the GFC and SDC whereas, during COVID-19, these contributors are rising NPLs, spikes in loan loss provisions, outdates liquidity frameworks, higher market volatility and higher lending risks.

Concentrating on systemic risk on a country level: Santis (2013) demonstrates an accurate example about the SDC, how Greece created contagion to countries with weak fiscal policies, higher dependency on foreign financing and low level of competitiveness.

De Bruyckere et al. (2013) are in alignment with Santis' (2013) statement, that countries with the highest debt-to-GDP ratio contribute most to contagion. Furthermore, Santis (2013) also explains about countries contributing to contagion due to weak economic foundations. During the SDC these countries were Spain, Belgium, Portugal, Austria, Ireland, France, and Italy. On the other hand, Aslam et al. (2021) assert that the contribution to contagion is the spillover during COVID-19 and Sweden and the Netherlands are the largest contributors to directional spillover while the least derives from Ireland and Poland. Countries that receive the most contagion are Belgium and Germany. Thus far, no real alignments on Santis' (2013) and De Bruyckere et al (2013) arguments were found concerning Aslam's et al. (2021) discovery. Lastly, D'Erman et al. (2020) affirms that EU member states must commit to a full collaboration with a reformed banking union. Here the statement made by D'Erman et al. (2020) is in alignment with Lagarde's (2021) and Gobbo's (2020) as precautions in the financial sector, against a new recession is vital for the European economy. To conclude, during the SDC the country contributing most to spillover is Greece due to its high-debt-GDP ratio, dependency on foreign aid, and weak fiscal policies. During COVID-19, large contributors are Sweden and the Netherlands. However, due to fact that not many studies have been conducted on spillover effect in Europe concerning COVID-19, it is soon to settle a verdict if causalities are in alignment with Santis (2013) and De Bruyckere et al. (2013).

2.10 Literature gaps

To answer this dissertation's main question "How resilient is the European economy during crisis periods based on systemic risk, and the volatility spillover between countries and economic sectors"; the following literature gaps will add to the existing literature described above. First, Current studies only observed data around the first wave. Observations on how systemic risk develops after distributions of vaccines can be a useful addition to this topic. Second, studies that observe systemic risk volatility spillover on a sectoral level in the Euro-zone are scarce: This dissertation will be an addition to this topic by filling this gap. Third, studies that observe systemic risk spillover on a country level on all member states of the EU are still scarce. Lastly, studies that observe daily sectoral stock, and country stock data in the EU

over a timeframe between 2006 till 2021 are still scarce. Therefore, this dissertation will be an addition to the existing literature by comparing three crisis scenarios based on systemic risk, and how the spillover effect puts forward across Euro-zone countries and economic sectors.

3. Methodology

3.1 Introduction

This section outlines the hypothesis of section one's research questions. Furthermore, its purpose is to inform the reader of the methodology that has been conducted to form a conclusion on Europe's economic resilience during times of crises by implementing models on measuring systemic risk, volatility spillover and connectedness on two dimensions namely: sectors and countries. In addition, this section will address the ethical consideration and the green status approval by managing secondary research from the Investing.com database. Lastly, limitations of this research will be addressed.

3.2 Research question and hypothesis

Research question one: What European countries and sectors contribute most to spillover transmission during COVID-19
Hypothesis one: During COVID-19, the largest contributors of spillover are the Netherlands and Sweden. During COVID-19 these sectors, based on the US and Chinese economy are Energy, industrials, materials, and utilities.

Research question two: What European countries and sectors are most exposed to spillover transmission during COVID-19
Hypothesis Two: Based on studies of the Chinese market concerning COVID-19, these sectors are healthcare and industrial goods and services. Countries most exposed to spillover are Belgium and Germany.

Research question three: in terms of volatility spillover, are there commonalities between the GFC, SDC and COVID-19 of volatility transmission channels on a country, and sectoral level?
Hypothesis three: The banking sector as a prominent spillover channel is a commonality of all crisis scenarios. Among countries, a commonality is Greece.

3.3 Research paradigm

The research paradigm most conform for this dissertation is positivism. Since this dissertation is based on scientific empiricist methods to process data and facts, results are unaffected by human perceptions or bias. Positivism also relates to causal relationships generated from data by implementing universal laws in order to forecast certain behavior (Saunders et al, 2019). Therefore, data sampling consists of large quantitates of stock market returns. Also, gathering daily percentages returns is imperative for quantitative procedures to generate results accurate as possible. In this dissertation, applications of time-series analysis are conducted. This quantitative approach often used in econometry is an effective method that presents how variables develop over time, and how variables can be correlated. Section 3.6 provides further explanation on applications of time-series analysis

3.4 Ethical considerations

All extracted data that is used for this dissertation is publicly available and therefore considered as secondary data. Therefore, ethical risk value is low. On June 21, 2021, the Northumbria university staff of the Amsterdam campus have granted the green status of the ethics approval submission.

3.5 Data collection

Sector data was gathered from Investing.com by using iShares STOXX Europe 600 Exchange Traded Funds (ETFs hereinafter) daily returns in percentages. Two data samples were assembled. One that combined the GFC and SDC of the period January 2, 2007, till December 28, 2012. The second sample regarding COVID-19 comes from February 3, 2020, till June 25, 2021. Specifications about what sectors this sample includes is listed on table 3.1

Sector	symbol
iShares STOXX Europe 600 Health Care	SXDPEX
iShares STOXX Europe 600 Industrial Goods & Services	SXNPEX
iShares STOXX Europe 600 Food & Beverage	SX3PEX
iShares STOXX Europe 600 Banks	SX7PEX
iShares STOXX Europe 600 Technology	SX8PEX
iShares STOXX Europe 600 Insurance	SXIPEX
iShares STOXX Europe 600 Oil & Gas	SXEPEX
iShares STOXX Europe 600 Utilities	SX6PEX
iShares STOXX Europe 600 Telecommunications	SXKPEX
iShares STOXX Europe 600 Construction & Materials	SXOPEX
iShares STOXX Europe 600 Basic Resources	SXPPEX
iShares STOXX Europe 600 Travel & Leisure	SXTPEX

Table 3.1: European sector ETFs

Country data was gathered from Investing.com and the Bloomberg database. Accordingly, daily country index returns in percentages have been gathered. Again, two data samples were assembled. One assembles the GFC and SDC which dates from June 1, 2007, till February 1, 2012. The second sample, belonging to COVID-19 comes from November 4, 2019, till June 25, 2021. One essential aspect to notice regarding this sample is the exclusion of one important country Portugal. Specifications of the data sample is listed on table 3.3 and table 3.4.

Country	Index
Austria	ATX
Belgium	BEL20
Denmark	OMXC20
Finland	OMXHPI
France	CAC 40
Germany	DAX
Greece	ATHEX
Ireland	ISEQ
Italy	FTSE MIB
Netherlands	AEX
Poland	WIG20
Spain	IBEX35
Sweden	OMXS30
UK	FTSE 100
Switserland	SMI
Iceland	OMXI10

Table 3.2: European stock indices

All missing data have been filled with percentage returns from the previous day to make all time stamps match. Finally, ETF and index returns are most suitable for this dissertation's quantitative approach. Studies from (Shu-Lien & Yun-Huan, 2019) on

return spillover between tourism ETFs, and Aslam et al. (2021) on volatility spillover among European markets, who conducted similar methodologies, demonstrate that Index and ETF returns generate sufficient results. Summary data of all sectors and countries that belong in this data sample can be found in table 3.3 and table 3.4

Summary data sectors	Sum	Average	Min	Max	Count	StDev
Healthcare	28.98	0.0154	-9.01	7.61	1876	1.161
Oil and Gas	1.54	0.00082	-17.08	15.78	1876	1.897
Basic Resources	86.74	0.0462	-14.1	18.24	1876	2.702
Banking	-58.58	-0.0312	-14.93	17.62	1876	2.402
Utilities	-39.98	-0.0213	-14.09	10.33	1876	1.472
Insurance	-16.52	-0.00881	-14.41	14.53	1876	2.145
Telecommunications	-22.21	-0.0118	-10.77	8.94	1876	1.368
Technology	38.46	0.0205	-9.78	9.87	1876	1.753
Food and Beverages	59.36	0.0316	-9.29	7.56	1876	1.157
Travel and leisure	139.58	0.0744	-15.08	10.21	1876	2.498
Industrial goods and services	51.14	0.0272	-12.03	10.57	1876	1.827

Table 3.3. Summary data of sectors. Note: All returns are percentages

Summary data countries	Sum	Average	Min	Max	Count	StDev
Spain	-33.59	-0.0207	-14.06	14.434	1621	1.837
Netherlands	-4.5	-0.00277	-10.75	10.548	1621	1.697
Ireland	-54.651	-0.0337	-13.032	10.222	1621	1.95
France	-21.942	0.0135	-12.28	11.176	1621	1.7888
Belgium	-40.624	-0.025	-13.8	9.659	1621	1.653
Austria	-33.88	-0.0209	-13.85	12.773	1621	2.079
Iceland	83	0.0512	-21.3	35.26	1621	1.663
Switzerland	15.236	0.0094	-9.24	9.911	1621	1.271
UK	-16.253	-0.01	-11.52	10.098	1621	1.63
Sweden	20.153	0.0124	-10.57	10.368	1621	1.747
Poland	-20.38	-0.0125	-13.28	8.496	1621	1.813
Italy	-54.044	-0.0333	-16.64	11.48	1621	1.901
Greece	-135.154	-0.0834	-13.39	14.374	1621	2.17
Germany	19.116	0.0117	-12.24	11.402	1621	1.723
Denmark	-26.913	-0.0166	-8.008	7.468	1621	1.216
Finland	5.839	0.0036	-10.23	8.303	1621	1.628

Table 3.4. Summary data of countries. Note: All returns are percentages

3.6 Methodology

The methodology is based on time series analysis. Time series analysis is conducted to analyze a series of data gathered over a specific time-interval. What makes time series analysis unique from other data analysis is that time series can present how variables develop over time. Therefore, time is an important factor as it

presents how data can change over the series of appointed data as well as end results. It ensures to add an extra source of intelligence, plus a series of dependencies between data. One condition to conduct time series analysis is to apply a large quantity of data to assure reliability and coherence. Consequently, an ideal sample size on time series analysis can carve noisy data, assure that patterns or trends are not deviations but instead, can expect season diversity. In addition, time series analysis can be applied to make forecasts based on the given data. Another important condition to conduct time series analysis is to use non-stationary time dependent data. Therefore, for economics, working with time series analysis is an excellent technique when, for example trends can be discovered based on stock market data (Tableau.com, 2020). This dissertation will apply the Dynamic Connectedness Approach based on DCC-GARCH (1,1) which is an extension of the GARCH model explained in section 2.4. It is strongly believed that this methodology is a breakthrough in econometrics as it determines how shocks impact over a pre-arranged network which demonstrates how the connectedness mechanism of previous financial crises spreads out via numerous avenues within the financial structure (Gabauer, 2020). According to Diebold and Yilmaz (2014), this method is useful as a warning mechanism for various economic applications. Therefore, observing how volatility transmits through other channels in the economy is a crucial application for investors and policymakers to recognize substantial transmission channels (Gabauer, 2020). A major benefit of the DCC-GARCH (1,1) compared with other models such as VAR is that it detects probabilities of diversity in conditional correlations throughout a given timeframe. On studies of financial markets, this a helpful aspect when for example, dynamic investor behavior in reaction to news and specific developments must be investigated. Furthermore, measuring dynamic conditional correlations is opportune to investigate how herding behavior in financial markets affect possible contagion through other avenues in the market (Celík, 2012). See also exemplary studies from Corsetti, et al. (2005), Boyer, et al. (2006), Chiang, et al. (2007) and, Syllignakis and Kouretas (2011). Another benefit of the DCC-GARCH (1,1) model is its ability to compute correlation coefficients of standardized residuals which narrates the direct effect of heteroscedasticity (Chiang, et al 2007). Because volatility gets fine-tuned by the proceeding, there is no prejudice from volatility regarding time varying correlations (Celík, 2012). DCC-GARCH (1,1) regulates correlations on time-varying volatility on a continuous basis. On that

account, DCC-GARCH (1,1) is regarded a preferable computation for correlations (Celík, 2012, Cho & Parhizgari, 2008). To investigate time-varying conditional volatility, the DCC-GARCH (1,1) model akin to Engle (2020) and Gabauer, (2020) and is adopted. A definition of the DCC-GARCH (1,1) can be found in Appendix E. Using models like the DCC-GARCH (1,1) model are often combined with programming applications such as R-Studios or Python. In this dissertation, computations are made with an online DCC-GARCH (1,1) application provided by Gabauer (2020).

3.7 Limitations

Some limitations of this dissertation must be addressed. The first limitation concerns missing values of the countries Greece, Iceland, Sweden, Poland, Denmark, Finland during the GFC and SDC periods. This gap can be mitigated with results from previous studies. The second limitation, concerning sector data, is the absence of the automotive, media, and chemicals-sector.

4. Results and analysis

4.1 Introduction

This section presents an analysis on the performance of European based sector, and country stock indices during three crisis situations, based on results of the DCC-GARCH model, to understand Europe's economic resilience during crisis periods. The following sections will describe sectoral performance during the GFC, SDC and COVID-19 (section 4.2), followed by country-level performance during the SDC and COVID-19 (section 4.3).

4.2 Results and analysis on sectors

Here, an analysis of sectors concerning all crisis scenarios will present.

Table 4.1 presents the average connectedness table of 16 European sectors during the GFC and SDC. The vertical column 'FROM' indicates in percentage how much on average each variable is receiving shocks from others. The horizontal column 'Contribution TO others' indicates how much in percentage each variable on average is transmitting shocks to others. The horizontal column 'NET directional connectedness' tells the difference what each variable is transmitting minus what this same variable is receiving from its own, plus other variables. The total connectedness index (TCI hereinafter) explains to what extent a shock from one variable impacts other variables on average (Gabauer, 2020). Here, the TCI indicates a high degree of 74.21 percent. Hence, on average 74.21 percent of a shock from one sector spills over to others.

The first large transmitter of shocks during the GFC and SDC is the basic resource sector which transmits on average 182.35 percent to other sectors. The basic resource sector transmits shocks mostly the oil and gas sector with 26.99 percent, followed by industrial goods and services with 21.77 percent. Thereafter, the utilities sector with 19.71 percent. The second large transmitter of shocks is the banking sector which transmits on average 133 percent to other sectors. The banking sector transmits shocks mostly to the insurance sector with 20.29 percent, followed by the telecommunications sector with 15.92 percent. Thereafter, the utilities sector with 14.84 percent. The third largest transmitter of shocks is the insurance sector which

transmits on average 108.38 percent to other sectors. The insurance sector transmits shocks mostly to the banking sector with 15.61 percent, followed by the utilities sector with 12.53 percent, and thereafter, the telecommunications sector with 12.22 percent. The three least contributors of shock transmissions are healthcare, food and beverage and travel & leisure transmitting on average 18.26 percent, 22.84 percent, and 38.46 percent respectively.

The three largest receivers of shock transmissions are as follows: The utilities sector, receiving on average 85.42 percent of shock transmissions of all industries ranks first. The utilities sector receives the most from the basic resource sector with an average shock transmission of 19.71 percent, followed by the banking sector with an average shock transmission of 14.84 percent and thereafter, the insurance sector with an average shock transmission of 12.53 percent. The second largest receiver of shock transmissions is the food and beverage industry which receives on average 83.89 percent of shock transmissions from all industries. The food and beverage sector receives most shock transmissions from the basic resource sector with an average shock transmission of 17.05 percent, followed by the banking sector with an average shock transmission of 11.51 percent and thereafter, the insurance industry from which it receives an average shock transmission of 11.15 percent. The third largest receiver of shock transmissions is the telecommunications sector which receives shock transmissions on average 83.56 percent from other industries. This industry receives most shock transmissions from the basic resource sector with an average of 16.83 percent, followed by the banking sector on which the sector receives 15.92 percent on average. Thereafter, the insurance company on which telecommunications receives shock transmissions of 12.22 percent on average. The least three receivers of shock transmissions are the travel and leisure, basic resource, and banking which receive an average shock transmission of 42.57 percent, 56.03 percent, and 71.20 percent respectively.

The three main net transmitters of shocks are as follows: First, the basic resource sector with an average of 126.32 percent. Second, the banking sector with an average of 61.8 percent. Third, the insurance sector with an average of 30.57 percent. These three sectors, as being the main net transmitters, influence other sectors more than being influenced by them. The three main net receivers of shocks

are the following sectors: First, healthcare with minus 64.99 percent; second, food and beverage with minus 61.05 percent; third, telecommunications with minus 44.95 percent. This means that the healthcare-, food and beverage-, and telecommunications industry are mostly driven by the European economy during the GFC and SDC.

GFC & SDC	Healthcare	Oil.and.Gas	Basic.Res	Banking	Utilities	Insurance	Telecommı	Technologı	Food.and.B	Travel.and.lei	Industrials	FROM
Healthcare	16.75	7.43	14.66	12.41	5.24	10.82	5.92	8.88	4.54	5.04	8.33	83.25
Oil.and.Gas	1.68	18.01	26.99	12.79	5.15	10.35	3.63	7.74	2.12	3.14	8.39	81.99
Basic.Resources	1.04	7.90	43.97	12.10	3.39	9.24	2.57	7.17	1.53	2.27	8.83	56.03
Banking	1.38	6.00	19.24	28.80	4.18	15.61	3.85	7.76	1.58	3.09	8.50	71.20
Utilities	2.00	8.79	19.71	14.84	14.58	12.53	5.52	7.03	2.81	3.78	8.41	85.42
Insurance	1.53	6.35	19.21	20.29	4.63	22.19	3.78	7.82	1.98	3.37	8.86	77.81
Telecommunications	2.60	7.02	16.83	15.92	6.33	12.22	16.44	8.80	2.76	3.44	7.64	83.56
Technology	1.65	6.10	19.02	13.14	3.31	10.01	3.59	26.55	1.92	3.86	10.85	73.45
Food.and.Beverages	3.58	7.25	17.05	11.51	5.56	11.15	4.85	8.32	16.11	5.90	8.72	83.89
Travel.and.leisure	1.33	3.28	7.87	6.64	2.31	5.78	1.88	5.32	1.61	57.43	6.54	42.57
Industrial.goods.and.services	1.47	6.25	21.77	13.36	3.73	10.67	3.03	10.33	1.99	4.58	22.83	77.17
Contribution TO others	18.26	66.36	182.35	133.00	43.82	108.38	38.61	79.18	22.84	38.46	85.08	TCI
NET directional connectedness	-64.99	-15.62	126.32	61.80	-41.60	30.57	-44.95	5.73	-61.05	-4.11	7.91	**74.21**

Table 4.1: Average connectedness table based on sectors during the GFC and SDC

Table 4.2 presents the average connectedness table of the same European sectors during the period from February 3, 2020, till June 25, 2021, which is during the COVID-19 pandemic. Here, the TCI indicates 80.47 percent which means there is 6.26 percent higher connectedness compared with the GFC and SDC.

The three most substantial contributors of shock transmissions are the following three sectors: The first sector is the banking sector which contribute to shock transmissions with an average of 135.46 percent to other sectors. The banking sector transmits shocks the most to the insurance sector with 20.08 percent. Second, oil and gas with 16.95 percent; third, to travel and leisure with 15.59 percent. The second most substantial contributor to shock transmissions is the oil and gas sector with an average of 128.20 percent of shock transmissions to other sectors. First, the oil and gas sector transmit most to the telecommunications sector with 15.41 percent; second, to the banking sector with shock transmissions of 15.02 percent and finally, to the insurance sector with shock transmissions of 14.48 percent. The third most substantial contributor of shock transmissions is the basic resource sector which contributes on average 118.24 percent of shock transmissions

to other sectors. First, the basic resource sector delivers most shock transmissions to the oil and gas sector with 14.19 percent; second, to the industrial goods and services sector with shock transmissions of 14.06 percent and third, to the technology sector with shock transmissions of 13.41 percent. The three least contributors of shock transmissions are the healthcare, food and beverage, and-utilities sector with average shock transmissions of respectively 23.24 percent, 37.22 percent and 42.93 percent.

The three sectors most subject to receiving shock transmissions are as follows: First, the food and beverage sector which receives on average 90.29 percent of shock transmissions from other industries. The first most substantial transmitter to this sector is the banking sector which contributes to 13.03 of shock transmissions. Second, the oil and gas sector which contributes to 12.60 percent of shock transmissions. Third, the travel and leisure sector which contributes to 12.41 percent of shock transmissions. The second sector must subject to receiving shock transmissions is the industrial goods and services sector which receives on average 88.44 percent of shock transmissions from other sectors. First, this sector receives most shock transmissions from the travel and leisure sector with 14.59 percent. Second, from the banking sector with 14.49 percent of shock transmissions. Third, from the basis resource sector with shock transmissions of 14.06 percent. The third sector most subject to receiving shock transmissions is the utilities sector which receives on average 85.63 percent of shock transmissions from other sectors. The utilities sector receives the most from the oil and gas sector with shock transmissions of 12.14 percent, followed by the banking sector with 10.94 percent of shock transmissions. And eventually, the basis resource sector from which it receives 10.49 percent of shock transmissions.

The three largest net transmitters of shocks are the following sectors: First, the banking sector with 64.40 percent of net transmissions; second, the oil and gas sector with 54.82 percent of net transmissions; third, the basic resource sector with net transmissions of 48.62 percent. During COVID-19 these European sectors appear to be the most resilient as they control the economy more than being driven by it instead. The three largest net receivers of shock transmissions are as follows: First, the healthcare sector which is a net receiver of minus 61.75 percent. Second,

the food and beverage sector which is a net receiver of minus 53.07 percent. Third, the telecommunications sector which is a net receiver of minus 44.08 percent. These sectors are the least resilient as they are strongly controlled by other sectors in the European economy.

Between the GFC, SDC and COVID-19 there are some commonalities when looking at contributors of shock transmissions. During all scenarios, the banking and basic resource sector both played a huge on contributing to shock transmissions. Shock transmissions from the basic resource sector during the GFC and SDC are 64 percent higher compared with COVID-19. Shock transmissions from the banking sector was 2.46 percent stronger during COVID-19 compared with the GFC and SDC. What sets the three crisis scenarios apart is that the oil and gas sector played a more prominent role on shock transmissions during COVID-19 compared with the GFC and SDC. On the other hand, the insurance sector played a more prominent role during the GFC and SDC in comparison with COVID-19

In terms of shock transmission receivers between all crisis scenarios, there are some commonalities. During all crisis scenarios, the utilities sector is a prominent receiver of shock transmissions, with 0.26 percent slightly higher during COVID-19. Another commonality between all scenarios is the food and beverage sector, with 6.4 percent higher on receiving shock transmissions during COVID-19. Finally, the healthcare sector is another commonality of both scenarios as net receiving remains around minus 60 percent. What sets the crisis scenarios apart in terms of receiving, is that the telecommunications sector played a more prominent role during the GFC and SDC, while the industrial goods and services sector were more prominent during COVID-19.

COVID-19	Healthcare	Oil and Gas	Basic Resou	Banking	Utilities	Insurance	Telecommun	Technology	Food and Be	Travel and leisu	Industrials	FROM
Healthcare	15.02	9.08	8.67	5.29	7.69	7.33	6.59	14.25	7.24	8.92	9.92	84.98
Oil.and.Gas	1.16	26.61	14.19	16.95	3.03	8.80	4.06	5.04	2.64	11.20	6.30	73.39
Basic.Resources	1.18	14.27	30.38	14.51	2.70	8.10	2.92	6.61	2.15	9.95	7.22	69.62
Banking	0.63	15.05	12.91	28.94	2.37	10.78	3.55	3.99	2.41	12.85	6.53	71.06
Utilities	3.89	12.14	10.49	10.94	14.37	9.42	6.01	9.23	5.21	9.88	8.42	85.63
Insurance	1.53	14.48	13.13	20.08	3.84	14.53	4.20	5.72	3.20	11.77	7.52	85.47
Telecommunications	3.14	15.41	10.87	15.08	5.67	9.63	12.96	6.59	4.51	9.07	7.06	87.04
Technology	3.64	10.59	13.41	9.48	4.68	7.36	3.62	20.77	3.40	13.06	9.98	79.23
Food.Beverages	4.37	12.60	9.83	13.03	6.20	9.35	5.73	7.85	9.71	12.41	8.92	90.29
Travel.and.leisure	1.22	12.08	10.67	15.59	2.61	7.79	2.59	6.77	2.79	30.01	7.88	69.99
Industrial.goods.and.services	2.48	12.50	14.06	14.49	4.13	9.08	3.68	9.72	3.70	14.59	11.56	88.44
Contribution TO others	23.24	128.20	118.24	135.46	42.93	87.63	42.96	75.79	37.22	113.70	79.75	TCI
NET directional connectedness	-61.75	54.82	48.62	64.40	-42.69	2.17	-44.08	-3.44	-53.07	43.71	-8.69	80.47

Table 4.2: Average connectedness table based on sectors during COVID-19

Figure 4.1 and **figure 4.2** presents a graphic overview of the dynamic net total directional connectedness which can also be seen on the lower horizontal column of figure 4.1 and figure 4.2. Here we can see how all sectors behave within a specified timeframe. All values above zero indicate that the sector is a net transmitter, while values below zero indicate that the sector is a net receiver. Due to the dynamic net total directional connectedness, we can clearly see that the oil and gas sector was clearly a net receiver during the GFC and SDC while during COVID-19 it is the complete opposite. The basic resource sector has been a net transmitter consistently during the GFC and SDC while during the COVID-19, there was a moment around January 2021 that it was net receiving. Another factor worth mentioning is the travel and leisure sector, which was a consistent net transmitter during COVID-19, whereas during the GFC and SDC this sector was a net transmitter between 2007 and 2010, but afterwards a net receiver. Last worth mentioning regarding the GFC and SDC, is how the banking and insurance sector were dominating all sectors, except the basic resource sector. This demonstrates the importance how a collapse of the financial sector can create spillover to other economic sectors.

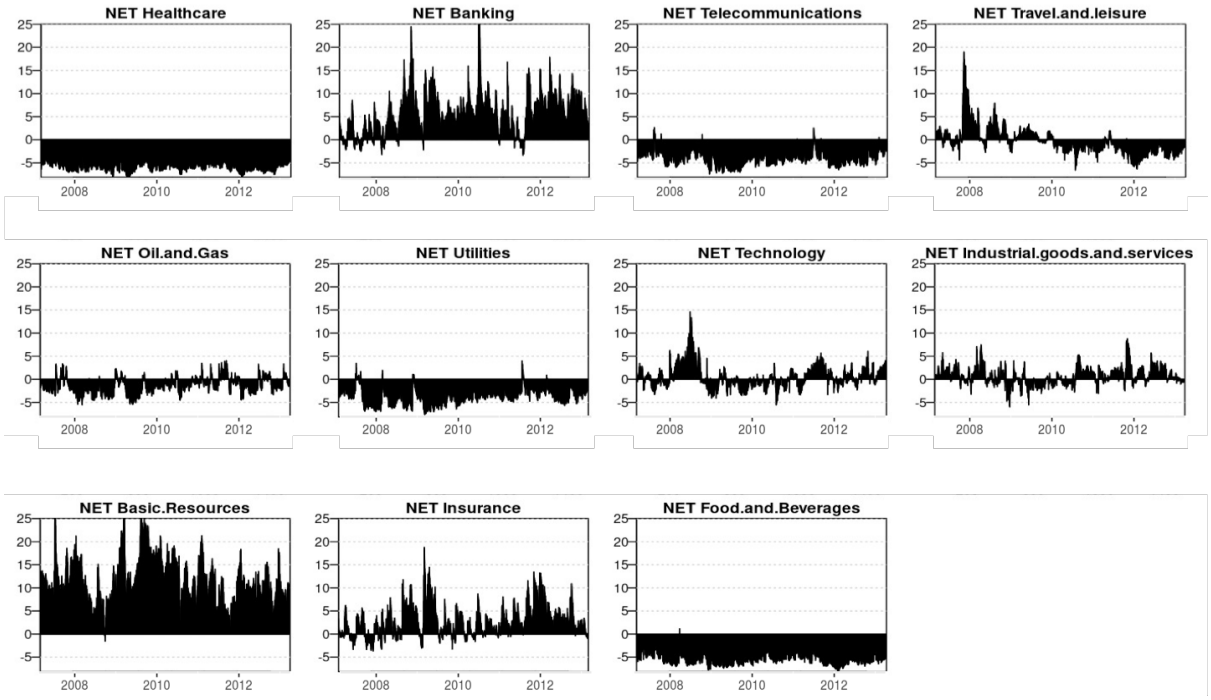


Figure 4.1: Dynamic net total directional connectedness based on sectors during the GFC and SDC

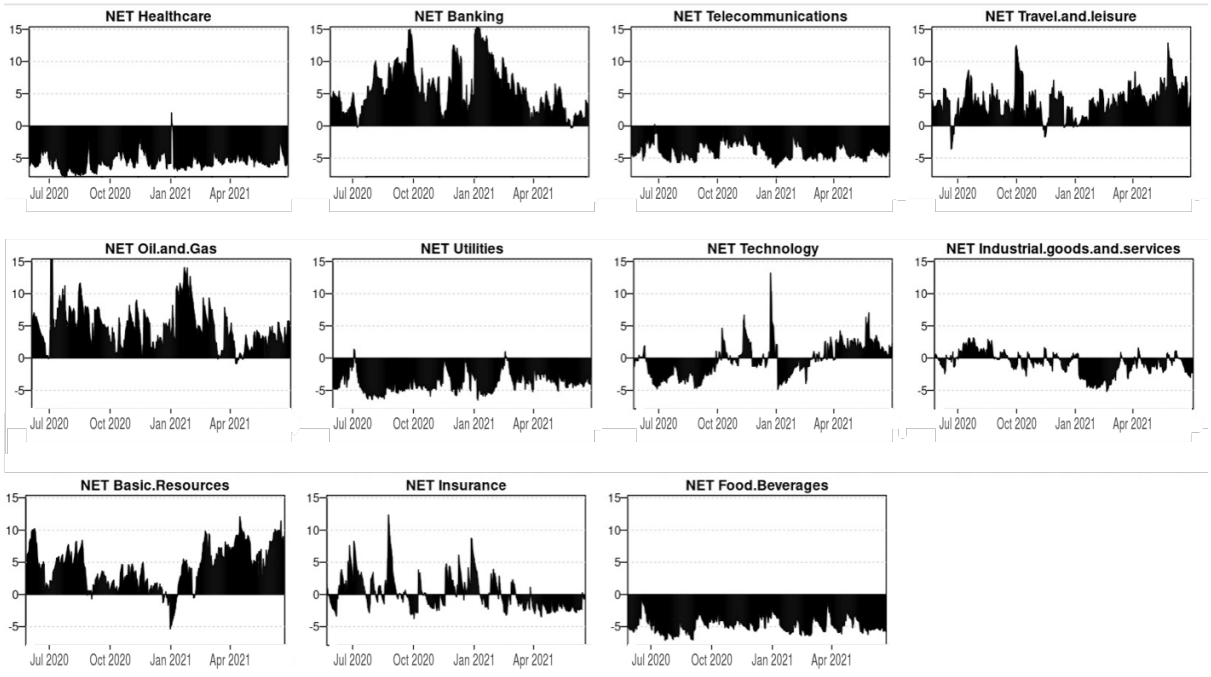


Figure 4.2: Dynamic net total directional connectedness based on sectors during COVID-19

4.3 Results and analysis on countries

Here, an analysis of countries concerning all crisis scenarios will present

Table 4.3 presents the average connectedness table based 16 on countries during the GFC and SDC. The TCI shows a high degree of connectedness with 76.66 percent which means that on average 76.66 percent of a shock from one country spills over to others.

During the GFC and SDC, the three most substantial contributors to shock transmissions are as follows: First, Austria which contributes to 115.76 percent of shock transmissions to other European countries. Most of these shock transmissions are sent to Denmark with 10.07 percent, followed by the Netherlands with 8.79 percent, and Sweden with shock transmissions of 8.75 percent. The second most substantial contributor to shock transmissions is Italy with an average of 113.87 percent to other countries. Most of these shock transmissions are sent to Spain with 10.59 percent, followed by Germany with shock transmissions of 9.68 percent, and the Netherlands with shock transmissions of 9.35 percent.
The third most substantial contributor to shock transmissions is Sweden with an average of 109.65 percent to other countries. Most of these shock transmissions are sent to Finland with 9.68 percent, followed by Switzerland with shock transmissions of 9 percent, and the Germany with shock transmissions of 8.98 percent. The three least contributors to shock transmissions are Iceland, France, and Denmark with shock transmissions of respectively 1.20 percent, 11.26 percent, and 29.11 percent.

Three countries that are most subject on receiving shock transmissions are as follows: First, Denmark which receives on average 92.33 percent of shock transmissions from other countries. Most of these shocks come from Austria which transmits 10.7 percent, followed by Ireland from which it receives 9.04 percent of shock transmissions, and Sweden which contributes to 8.30 of shock transmissions. The second country most subject on receiving shock transmissions is Switzerland which receives shock transmissions of 91.39 percent on average. Most of these shocks come from Sweden with shock transmissions of 9 percent, followed by Ireland with shock transmissions of 8.64 percent, and Italy from which it receives 8.40 percent of shock transmissions. The third country most subject on receiving

shock transmissions is Belgium which receives shock transmissions of 90.63 percent on average. Most of these shock transmissions come from Italy from which the Netherlands receives 9.49 percent, followed by Austria with 8.86 percent of shock transmissions, and Spain which contributes to 8.63 percent of shock transmissions.

Three of the largest net transmitters during the GFC and SDC are the following countries: First, Austria with a net transmission of 32.56 percent; second, Italy with a net transmission of 27.10 percent; third, Ireland with a net transmission of 25.01 percent. These three countries were mostly rather controlling other European countries during the SDC, than being controlled by other countries instead. Three countries mostly controlled by other countries are the net receivers. These countries are Denmark, the Switzerland and Iceland with a net receiving of respectively minus 63.21 percent, minus 54.74 percent and minus 22.42 percent.

GFC & SDC	Spain	Netherlands	Ireland	France	Belgium	Austria	Iceland	Switzerland	UK	Sweden	Poland	Italy	Greece	Germany	Denmark	Finland	FROM
Spain	13.17	7.26	7.63	0.85	6.34	8.73	0.07	2.59	5.60	8.15	6.39	10.59	6.55	8.12	1.85	6.10	86.83
Netherlands	8.35	9.64	8.04	0.82	6.75	8.79	0.07	2.94	6.50	8.77	6.43	9.35	5.81	8.51	2.21	7.03	90.36
Ireland	7.20	6.65	18.54	1.02	5.78	8.57	0.09	2.86	5.58	8.05	6.11	8.06	6.07	6.86	2.34	6.24	81.46
France	0.58	0.52	0.79	93.96	0.35	0.55	0.00	0.22	0.36	0.52	0.32	0.51	0.48	0.41	0.07	0.35	6.04
Belgium	8.63	7.92	8.28	0.71	9.37	8.86	0.07	2.95	5.84	8.38	6.03	9.49	6.26	8.07	2.28	6.88	90.63
Austria	7.70	6.69	7.89	0.80	5.74	16.79	0.07	2.56	5.23	8.08	6.54	8.33	7.50	7.19	2.43	6.45	83.21
Iceland	1.51	1.28	2.94	0.73	1.42	1.84	76.38	0.64	2.45	1.71	1.59	1.39	1.47	1.40	0.77	2.50	23.62
Switzerland	7.36	7.35	8.64	0.92	6.27	8.41	0.06	8.61	6.88	9.00	6.03	8.40	5.16	7.92	2.26	6.71	91.39
UK	7.96	8.16	8.50	0.77	6.24	8.54	0.12	3.47	11.03	8.49	6.19	8.89	4.42	8.38	2.21	6.64	88.97
Sweden	7.68	7.25	8.10	0.82	5.89	8.75	0.07	2.96	5.59	14.65	6.70	8.47	5.21	7.95	2.16	7.74	85.35
Poland	7.08	6.19	7.07	0.71	4.97	8.21	0.08	2.28	4.75	7.82	21.75	7.57	7.22	6.63	2.01	5.68	78.25
Italy	9.72	7.45	7.74	0.71	6.42	8.64	0.08	2.70	5.68	8.21	6.24	13.23	6.40	8.24	2.00	6.54	86.77
Greece	6.16	5.01	6.29	0.85	4.45	8.27	0.07	1.83	3.25	5.52	6.60	6.58	33.78	4.99	2.04	4.31	66.22
Germany	8.75	7.94	7.77	0.64	6.41	8.80	0.08	2.96	6.24	8.98	6.45	9.68	5.57	10.85	2.02	6.86	89.15
Denmark	6.61	6.98	9.04	0.21	6.13	10.07	0.11	2.89	5.60	8.30	6.59	7.97	7.30	6.85	7.67	7.68	92.33
Finland	7.29	7.31	7.74	0.71	6.11	8.74	0.14	2.79	5.43	9.68	6.05	8.60	5.34	7.68	2.47	13.93	86.07
Contribution TO others	102.57	93.94	106.47	11.26	79.27	115.76	1.20	36.65	74.97	109.65	84.27	113.87	80.76	99.18	29.11	87.71	TCI
NET directional connectedness	15.75	3.58	25.01	5.22	-11.36	32.56	-22.42	-54.74	-14.00	24.29	6.02	27.10	14.54	10.03	-63.21	1.64	76.66

Table 4.3: Average connectedness table based on countries during the GFC and SDC

Table 4.4 presents the average connectedness table based on 16 European countries during the COVID-19 pandemic. Here, the TCI indicates 85.67 percent which means, again a degree of connectedness although, 0.81 percent less compared with the SDC.

During COVID-19, the most substantial contributors of shock transmissions on a country level are the following: First, Germany which contributes to an average shock transmission of 118.96 percent. Most of Germany's shock transmissions are

sent to Switzerland with 9.09 percent, followed by Italy with shock transmissions over 9.05 percent, and the Netherlands with shock transmissions of 8.88 percent. The second most substantial contributor of shock transmissions is Italy which contributes to 114.02 percent on average. Most of Italy's shock transmissions are sent to Germany with 8.89 percent, followed by Spain with shock transmissions of 8.82 percent, and France with shock transmissions of 8.79 percent. The third most substantial contributor of shock transmissions is Belgium with an average of 112.36 percent. Most of Belgium's shock transmissions are sent to Spain with 8.38 percent, followed by Austria with shock transmissions of 8.35 percent, and France with shock transmissions of 8.25 percent. Countries that contribute the least to shock transmissions are Iceland, Switzerland and Denmark which respectively estimate on average 22.50 percent, 41.60 percent, and 38.85 percent to other countries.

Three European countries most subject to receiving shock transmissions are as follows: First, Switzerland receives the most shock transmissions from other countries with an average of 92.36 percent. Most of these shocks come from Germany with 9.09 percent, followed by Italy with shock transmissions of 8.19 percent, and Sweden with shock transmissions of 7.77 percent. The second country that receives most of the shock transmissions is France with an average of 91.05 percent from other countries. Most of these shock transmissions come from Spain with 8.85 percent, followed by Italy with shock transmissions of 8.79 percent, and Germany with shock transmissions of 8.77 percent. The third country most subject to shock transmissions from other countries is the Netherlands which receives shock transmissions of 90.71 percent from other countries. Most of these shock transmissions come from Germany with 8.88 percent, followed by Italy with shock transmissions of 8.10 percent, and Belgium with shock transmissions of 8.07 percent.

Three of the largest net transmitters during COVID-19 are as follows: First, Germany with a net transmission of 30.85 percent; second, Italy with a net transmission of 26.02 percent; third, Belgium with a net transmission of 25.38 percent. These three countries were mostly controlling other European countries during COVID-19, than being controlled by other countries instead. Three countries most controlled by other countries are net receivers. These countries are Iceland, Switzerland and Denmark

with a net receiving of respectively minus 54.67 percent, minus 50.76 percent and minus 44.90 percent.

In terms of net transmitters on a country level, Spain, Italy, and Austria are the largest net transmitters during the GFC and SDC whereas, during COVID-19 the largest net transmitters are Italy, Germany, and Belgium, which makes Italy a common net transmitter of both scenarios. The largest net receivers during the GFC and SDC are Denmark, Switzerland, and Iceland, while during COVID-19 these countries are Iceland, Switzerland, and Denmark. Both crisis scenarios have Iceland and Switzerland as prominent net receiver in common.

COVID-19	Spain	Netherlands	Ireland	France	Belgium	Austria	Iceland	Swiss	UK	Sweden	Poland	Italy	Greece	Germany	Denmark	Finland	FROM
Spain	12.60	5.38	7.28	7.81	8.38	8.66	1.41	2.21	5.86	5.87	6.20	8.82	5.66	7.85	1.55	4.44	87.40
Netherlands	6.99	9.29	6.76	7.30	8.07	6.45	1.43	3.52	6.43	7.03	5.94	8.10	5.10	8.88	3.48	5.24	90.71
Ireland	8.04	5.79	14.71	6.83	8.00	7.34	1.77	2.37	6.17	6.32	5.88	7.77	4.50	7.86	2.18	4.50	85.29
France	8.85	6.35	6.96	8.95	8.25	7.47	1.37	2.94	6.29	6.47	5.87	8.79	5.66	8.77	2.23	4.77	91.05
Belgium	8.22	6.12	7.08	7.15	13.02	7.84	1.44	2.50	5.99	6.31	6.09	7.99	5.15	8.21	1.98	4.91	86.98
Austria	9.05	5.18	6.94	6.89	8.35	13.55	1.46	2.01	5.71	5.87	6.58	8.24	6.20	7.78	1.33	4.84	86.45
Iceland	5.86	4.54	6.55	5.10	6.13	5.83	22.83	1.92	4.62	5.86	6.29	5.43	6.60	6.06	2.21	4.17	77.17
Swiss	6.17	7.63	5.96	7.30	7.12	5.37	1.29	7.64	6.42	7.77	5.71	8.19	3.97	9.09	5.04	5.34	92.36
UK	7.87	6.65	7.49	7.48	8.20	7.34	1.50	3.08	9.94	6.75	6.07	7.59	5.13	7.62	2.34	4.94	90.06
Sweden	6.80	6.27	6.58	6.64	7.48	6.49	1.64	3.20	5.82	12.36	6.17	7.70	5.06	8.39	3.07	6.32	87.64
Poland	6.80	5.02	5.79	5.74	6.91	6.89	1.67	2.24	4.95	5.80	21.24	6.71	5.89	7.10	2.52	4.73	78.76
Italy	8.61	6.09	6.85	7.59	7.97	7.70	1.28	2.85	5.51	6.50	5.96	12.00	5.12	9.05	2.13	4.78	88.00
Greece	6.28	4.38	4.46	5.60	5.88	6.60	1.74	1.58	4.22	4.81	5.96	5.83	31.30	6.21	1.38	3.76	68.70
Germany	7.54	6.56	6.81	7.44	8.04	7.16	1.39	3.10	5.43	6.95	6.18	8.89	5.30	11.62	2.71	4.88	88.38
Denmark	4.41	7.63	5.63	5.72	5.78	3.66	1.54	5.12	5.02	7.65	6.58	6.34	3.53	8.16	18.25	4.98	81.75
Finland	6.90	6.30	6.28	6.58	7.82	7.21	1.56	2.95	5.73	8.49	6.69	7.63	5.23	7.94	2.70	9.99	90.01
Contribution TO others	108.38	89.90	97.43	101.18	112.36	102.02	22.50	41.60	84.17	98.45	92.17	114.02	78.09	118.96	36.85	72.61	TCI
NET directional connectedness	20.98	-0.81	12.14	10.14	25.38	15.58	-54.67	-50.76	-5.88	10.81	13.41	26.02	9.38	30.58	-44.90	-17.40	85.67

Table 4.4: Connectedness table based on countries during COVID-19

Figure 4.3 Shows Ireland's role when its property sector collapsed in 2008. Ireland ranks number four of most prominent shock transmitters during the GFC and SDC. Since no tremendous shock transmitting is detected before 2009, it can be said that, according to measurements from the DCC-GARCH (1,1) model that Ireland was partly responsible for further shock transmissions. After 2009, Austria, Italy, Poland Sweden, and Germany started experiencing the aftermath of the GFC. This became more apparent when Spain and Greece created larger volumes of shock spillover through the rest of Europe. After 2010 countries that were most affected due to receiving shock spillover were the UK, Belgium, Iceland, Finland, Denmark, Poland, Sweden, Switzerland, and the Netherlands.

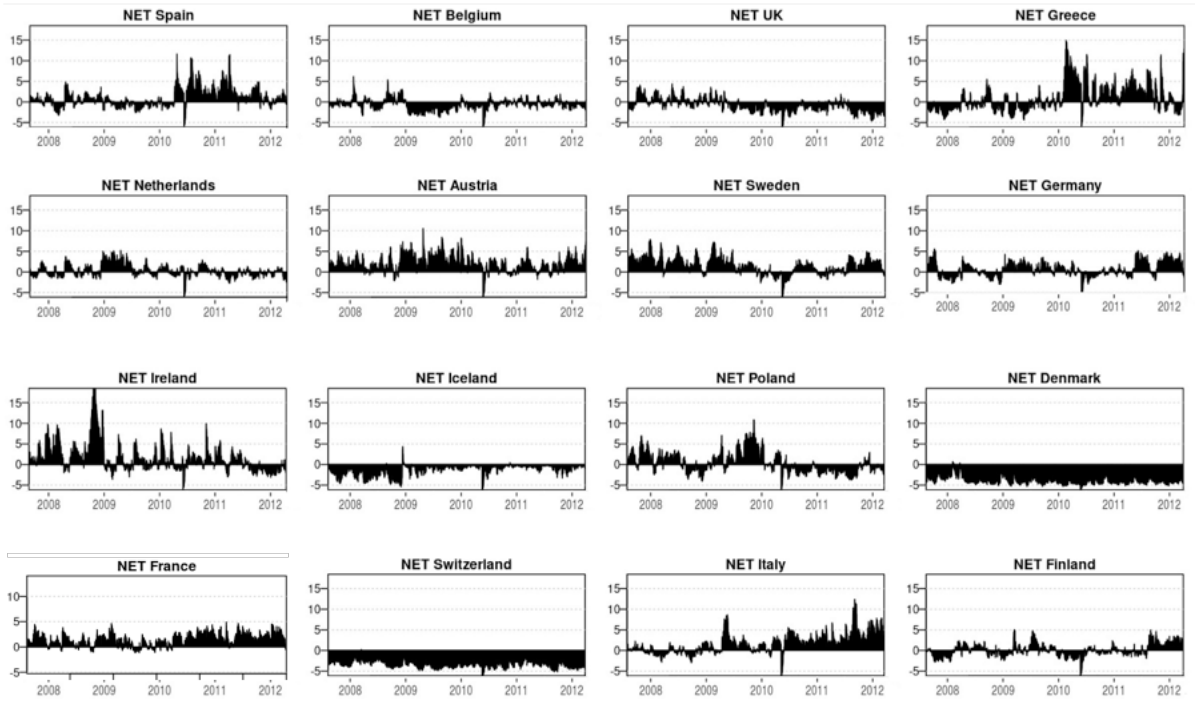


Figure 4.3: Dynamic net total directional connectedness based on countries during the SDC

By looking at **figure 4.4**, we will start to assess each country based on the number of COVID-19 cases. All data regarding COVID-19 countries have been obtained by Statista.com (2021) France with over 6.7 million cases shows some mild peaks of net transmitting from the first lockdown, while after the second lockdown it shows its largest peak of 7 percent of net transmitting. Afterwards, between March and June 2021, France appears to be less volatile towards other countries. The UK with cases over 6.7 million appears to be the least volatile as it was shaking between net receiving and net transmitting. Spain with cases over 4.8 million was heavily transmitting between September 2020 and May 2021. Spain's highest peak of net transmitting was during its second lockdown around spring 2021. Italy, with COVID cases over 4.5 million is one of the first countries affected by the pandemic. Italy clearly shows, that from the start of COVID-19 in Europe, it was heavily net transmitting. Afterwards, net transmitting kept swinging one and five percent. Germany, with COVID cases over 3.9 million was transmitting from the start with a rising trend, but clearly decreased volatility after vaccine distributions. Poland, with COVID cases over COVID cases of 2.8 million, appears to be net receiving during first lockdown measures, and increased net transmitting in a rising trend after many countries initiated their second lockdown. Vaccine distributions did not seem to

decrease Poland's volatility towards other countries. Netherlands, with over 1.9 million COVID cases is one the least volatile countries. Around second lockdown measures there was a small peak of 5 percent net transmitting. Afterwards, around the distributions of vaccines the Netherlands was more net receiving. Belgium with over 1.2 million cases shows high net transmitting of volatility during first and second lockdown measures. After vaccine distributions Belgium seems to decrease volatility to other countries. Sweden with 1.1 million COVID cases did not seem to cause much volatility during first and second lockdown periods. However, volatility seems to increase in summer 2021.

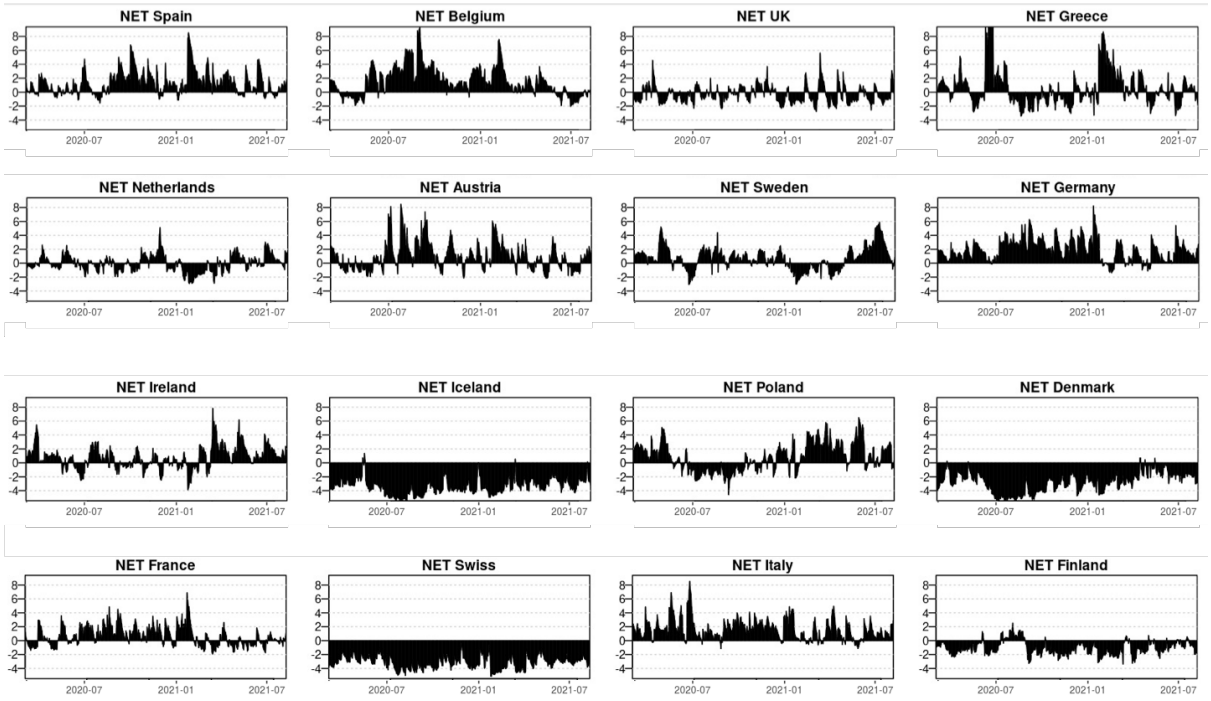


Figure 4.4: Dynamic net total directional connectedness based on countries during the COVID-19

5. Conclusions

5.1 Introduction

This section intends to integrate all results that were based upon this dissertation's three main questions, with all related theories while recognizing any discord that have been encountered throughout this research, followed by an explanation to the main question. In the end, and evaluation of the literature, limitations and recommendations are presented

5.2 Answer to research questions

Research question one: What European countries and sectors contribute most to spillover transmission during COVID-19:

Answer to research question one: The banking, oil and gas, and basic resource -sector are the largest contributors of spillover during COVID-19. On a country level largest contributor are Germany, Italy, and Belgium

Research question two: What European countries and sectors are most exposed to spillover transmission during COVID-19?

Answer to research question two: Iceland, Switzerland, and Denmark are the countries most exposed to spillover during COVID-19. The healthcare, food and beverage, and the telecommunications are most exposed to spillover during COVID-19

Research question three: in terms of volatility spillover, are there commonalities between the GFC, SDC and COVID-19 of volatility transmission channels on a country, -and sectoral level?

Answer to research question three: During all crisis scenarios, both the banking and basic resource sector are the largest net transmitters. Basic resource was net transmitting 126.32 percent during the GFC and SDC, while this was 48.62 percent during COVID-19. Banking was net transmitting 61.80 percent during the GFC and

SDC, and 64.40 percent during COVID-19. These sectors appear to be the most influential during economic downturns.

In terms of receiving, commonalities are the food and beverage sector which was net receiving with minus 61.05 percent during the GFC and SDC, and minus 53.07 during COVID-19. The utilities sector was net receiving with minus 41.60 percent during the SDC and GFC, and minus 42.69 of net receiving during COVID-19. Finally, the healthcare sector was net receiving with minus 64.99 percent during the SDC and GFC, and minus 61.75 percent during COVID-19. These sectors appear to be the most subject of volatility from other sectors during economic downturns. Regarding net transmitters on a country level, Spain, Italy, and Austria are the largest net transmitters during the GFC and SDC while, during COVID-19 the largest net transmitters are Italy, Germany, and Belgium, making Italy a common net transmitter of both crisis scenarios. The largest net receivers during the GFC and SDC are Denmark, Switzerland, and Iceland, whereas during COVID-19 these countries are Iceland, Switzerland, and Denmark. Both crisis scenarios have Iceland and Switzerland in common as net receiver.

Main question: How resilient is the European economy during crisis periods based on systemic risk, and the volatility spillover between countries and economic sectors?

Answer to main question: The DCC-GARCH (1,1) model measures increased connectedness during COVID-19 in comparison with the GFC and SDC. On a country level, TCI indicates 85.76 percent, while on sectoral level TCI indicates 80.74 percent during COVID-19. On the other hand, during the GFC and SDC, TCI on country-level indicates 76.66 percent, and 74.21 percent on sectoral-level. The higher the TCI, the more we speak of systemic risk. COVID-19 shows there is a higher level of connectedness so implementing the right measures in a timely manner can prevent the European economy from further collapse. Banks have always been at the frontlines during times of crises and the banking sector always seems to contribute large percentages of spillover transmissions. Which is why preventive measures should always be devoted to the financial sector. On a country-level, countries with higher debt-GDP ratios are often contributors to spillover to

other countries. Especially if the geographical distance is closer, and, if the country is a member of an integrated economy such as the European Union. To answer this dissertation's main question: The European economy is less resilient in terms of systemic risk and volatility spillover which is caused by high-debt GDP ratios from countries such as Italy, Austria, Spain, and Ireland. The SDC is a fine example in that regard. Also, banks contribute to systemic risk mainly because of NPLs or weak capital buffers, or both, and are concerns that must be high on the ECB's agenda. Lastly, the methodology of this research does not extend in the field of NPLs, capital buffers and debt-GDP ratios, but this could be a suggestion for future research with the focus on countries with large spillover transmission.

5.3 Re-evaluation of literature

Theories featuring the GARCH model brought perspective of how studies by Diebold and Yilmaz (2014), Cho and Parhizgari (2008), Corsetti et al. (2005), Celík (2012) and Gabauer (2020) measure volatility spillover channels among foreign exchange markets, countries, or economic sectors. Since GARCH models are often used by econometrists and risk managers to find measure volatility and connectedness between specified variables, it demonstrates to be an appropriate methodology to examine how COVID-19 impact volatility spillover among sectors and countries. After re-evaluation of all related literature, it becomes noticeable that systemic risk does not only consist of failing financial institutions as we know from the GFC, but also unexpected events such as the COVID-19 pandemic. This argument can be affirmed by Trichet (2009). Furthermore, according to Fouque and Langsam (2013), COVID-19 increases systemic risk since it impacts the economy, which eventually, decreases capital flows and creates negative GDP growth, and decreases capital flows. This argument is supported by Ortman and Tripier (2021) who projected a governmental deficit of 0.06 percent of GDP to 8 percent of GDP. This argument can be supportive to investigate if there is indeed a strong correlation between large transmitters such as Germany, Italy and Belgium, and country's debt-GDP ratio. An argument by Liu, Hong, & Wang (2015) brings attention on monitoring and controlling financial risk by assessing fluctuations. These fluctuations cultivate spillover from country to country or contagion from one sector to another, often caused by regulatory reforms or natural disasters. The argument by Liu et al, (2015)

gave perspective to monitor spillover effect on a sectoral, and country level. Based on these arguments, this dissertation is to present a comparison of how spillover channels differentiate between the COVID-19 pandemic and previous financial crisis scenarios, on both country, and sectoral level. Hence, findings of this dissertation are how the insurance sector was a prominent contributor to spillover during the GFC and SDC, while the oil and gas sector was a more prominent contributor of spillover during COVID-19. With this information investors and policymakers can anticipate the market by monitoring other channels related to these sectors. De Bruyckere at al. (2013) who studied risk contagion during the SDC, suggest that banks with weak capital buffers, fragile funding structure and less conservative banking operations, are prone to spillover. On a country level, de Bruyckere et al (2013) suggest that countries with high debt ratios are often responsible for spillover. Santis (2013) concludes that Greece played a huge role on creating spillover to countries with weak fiscal policies, higher dependency on foreign financing and low level of competitiveness. Findings of this dissertation support the arguments by Santis (2013) and de Bruyckere et al (2013) by indicating that the banking and insurance sector contribute to 133 percent, and 108 percent respectively to nine other sectors of the European market. However, Santis (2013) did not mention Ireland's role about its collapse of the property sector in 2008. Findings of this dissertation ranks Ireland number four as prominent net transmitter of volatility with 25 percent. Greece's net transmitting reaches only 14 percent of volatility. The reason behind this is Ireland's tremendous shock transmission around winter in 2008, which created spillover through the rest of Europe. Austria, Italy, Poland, Sweden, and Germany were starting to experience the aftermath of the GFC in 2009. Around 2010 the aftermath of the GFC became more apparent as Spain and Greece created large volumes of shock spillover through the rest of Europe. After 2010, countries that were heavily afflicted by the SDC were the UK, Belgium, Iceland, Finland, Denmark, Poland, Sweden, Switzerland, and the Netherlands. Hence, findings of this dissertation sheds light that Greece indeed played a huge role, along with other countries, but also suggests that the collapse of the property sector before 2009 contributed to further systemic risk. Moreover, findings of this dissertation are supportive to Santi's (2013) arguments of how the SDC is related to country-specific risk, contagion risk, and spillover transmission. A suggestion by Shahzad et al. (2021) explains to pay close attention to four sectors namely, industrials, energies, materials, and utilities

since these can destabilize the market. Findings by Shahzad et al. (2021) are based on the Chinese market and conclude that disastrous events like COVID-19 have a severe impact on Chinese economic sectors. This dissertation's findings, based on European sectors, support the same discovery mainly on utilities and industrials. Another study based on the Chinese market by Wang-and Li (2021) discovered that the oil industry during 2003 and 2019 has always been volatile. Findings of this dissertation can relate since the oil and gas sector was one of the main contributors to spillover in Europe. Another study based on the US by Laborda and Olmo (2021) discovered that three sectors, technology, banking and insurance are the primary sectors that contribute to spillover transmission. This dissertation made the same findings, except for the technology sector which was mostly receiving spillover transmission. Aslam et al. (2021) researched 12 European countries between December 2019 and May 2020 and discovered that Sweden and the Netherlands contribute most to spillover, while the least originate from Ireland and Poland. This dissertation disagrees and instead, Germany, Italy and Belgium contribute most to spillover. However, the Netherlands and Sweden do contribute to a large amount of spillover, but they do not compare with the other countries mentioned.

5.4 Limitations and recommendations

Several limitations of this dissertation must be accepted. In terms of missing data: Countries such as Portugal, Bulgaria, Romania, and sectors such as automotive, materials and energy are absent in this data sample. Also, sampling data before the COVID-19 period could be useful to compare TCIs between tranquil and turbulent periods. Another limitation concerns the topic itself: To investigate more about Europe's resilience during crisis periods, investigating spillover channels would not be sufficient. This dissertation did not bring attention to country's debt-GDP ratios, banks capital buffers, or NPL's which are also prime indicators to measure an economy's strength. These topics are suggestive for further investigation.

References

Acharya, V., Pedersen, L., Philippon, T., & Richardson, M. (2016). Measuring Systemic Risk. *The Review of Financial Studies, Volume 30, Issue 1, January 2017, Pages 2–47*.

Adrian, T., & Brunnermeier, M. (2016). CoVaR. *The American Economic Review. Vol 106*, 1705-1741.

Akshay, J. (2018, November 30). *Veristrat.com*. Retrieved from Value at Risk (VAR)- Definition, Calculation Methodologies : https://www.veristrat.com/blog-valuation/value-at-risk-var/

Amar, A., Bélaïd, F., Youssef, A., & Guesmi, K. (2020). Connectedness among regional financial markets in the context of the COVID-19. *Applied Economic Letters*, https://doi.org/10.1080/13504851.2020.1854434.

Anderson, R. M., Heesterbeek, H., Klinkenberg, D., & Hollingsworth, T. D. (2020). How will country-based mitigation measures influence the course of the COVID-19 epidemic? *The Lancet, 395(10228), 931–934, https://doi.org/101016/S0140-6736(20)30567-5*.

Arellano, C., Bai, Y., & Mihalache, G. (2020). Deadly debt crises: COVID-19 in Emerging Markets. *National Bureau of Economic Research*.

Ari, A., Chen, S., & Ratnovski, L. (2020). COVID-19 and non-performing loans: lessons from past crises. *European Central Bank*.

Aslam, F., Ferreira, P., Mughal, K., & bashir, B. (2021). Intraday Volatility Spillovers among European Financial Markets during COVID-19. *International Journal of Financial Studies*, https://doi.org/10.3390/ijfs9010005.

Bernanke, B. (2012). *Lecture 3: The Federal Reserve's Response to the Financial Crisis," Chairman Bernanke's College Lecture Series*.

Bernhardt, D., & Eckblad, M. (2013, November 19). *Stock Market Crash of 1987*. Retrieved from FederalReserveHistory.com: https://www.federalreservehistory.org/essays/stock-market-crash-of-1987

Binswanger, H., & Rosenzweig, M. (1986). Behavioural and material determinants of production relations in agriculture. *The Journal of Development Studies. Volume 22, 1986. Issue 3*, https://doi.org/10.1080/00220388608421994.

Boyer, B., Kumagai, T., & Yuan, K. (2006). How do crises spread? Evidence from accessible and inaccessible stock indices. *Journal of Finance: 61*, 957-1003.

Brownlee, J. (2019, August 21). *Machinelearningmastery.com*. Retrieved from How to Model Volatility with ARCH and GARCH for Time Series Forecasting in Python: https://machinelearningmastery.com/develop-arch-and-garch-models-for-time-series-forecasting-in-python/

Bruyckere, V., Gerhardt, M., Schepens, G., & Vennet, R. (2013). Bank/sovereign risk spillovers in the European debt crisis. *Journal of Banking & Finance*, 4793-4809.

C Lagarde. (2021). *ESRB Annual Report 2020.* Frankfurt am Main, Germany: ESRB.

Casu, B., Girardone, C., & Molyneux, P. (2015). *Introduction to banking: Second edition.* Pearson.

Celík, S. (2012). The more contagion effect on emerging markets: The evidence of DCC-GARCH model. *Economic Modelling: 29*, 1946-1959 doi:10.1016/j.econmod.2012.06.011.

Chiang, Jeon, T., & Li, H. (2007). Dynamic correlation analysis of financial contagion: evidence from Asian markets. *Journal of International Money and Finance*, 1026-1228.

Cho, J., & Parhizgari, A. (2008). East Asian financial contagion under DCC-GARCH. *International Journal of Banking and Finance: 6*.

Choudhry, M. (2006). *An Introduction to Value-at-Risk: Fourth Edition.* John Wiley & Sons, LTD.

Control, E. C., & (ECDC). (2020). COVID-19 Situation update worldwide. Retrieved April 4, 2020, from https://www.ecdc.europa.eu/en/geographical-distribution-2019-ncov-cases.

Corsetti, Giancarlo, Pericoli, Marcello, Sbracia, & Massimo. (2005). 'Some contagion, some interdependence': More pitfalls in tests of financial contagion. *Journal of International Money and Finance: Vol 24(8)*, 1177-1199.

Crotty, J. (2009). Structural causes of the global financial crisis: a critical assessment of the 'new financial architecture. *Cambridge Journal Of Economics*, p 563–580.

D'Erman, V., Schure, P., & Verdun, A. (2020). Introduction to "Economic and Financial Governance in the European Union after a decade of Economic and Political Crises.". *Journal Of Economic Policy Reform, 23:3,*, 267-272. DOI.org/10.1080/17487870.2020.1762599.

Diebold, F., & Yilmaz, K. (2014). On the network topology of variance decompositions: Measuring the connectedness of financial firms. *Journal of Econometrics, 182*, 1119-134.

ECDC, & Control, E. C. (2020). *COVID-19 Situation update worldwide*. Retrieved from https://www.ecdc.europa.eu/en/geographical-distribution-2019-ncov-cases

Engle, R. (2020). Dynamic conditional correlation: A simple class of multivariate generalized autoregres- sive conditional heteroskedasticity models. *Journal of Business and Economic Statistics: 20*, 339-350.

Erica. (2021, April 15). *Aptech.com*. Retrieved from Introduction to the Fundamentals of Vector Autoregressive Models: https://www.aptech.com/blog/introduction-to-the-fundamentals-of-vector-autoregressive-models/

FCO (The Foreign & Commonwealth Office). (2020). *Foreign Secretary advises all British travelers to return to the UK now. Retrieved March 31, 2020*. Retrieved from https://www.gov.uk/government/news/foreign-secretary-advises-all-brit-

Ferreira, P., Aslam, F., Shahzad, K., & Bashir, B. (2020). Intraday Volatility Spillovers among European Financial Markets during COVID-19. *International Journal of Financial Studies*, https://doi.org/10.3390/ijfs9010005.

Fouque, J., & Langsam, J. (2013). *Handbook of Systemic Risk.* Cambridge University Press.

FSB. (2011). Retrieved from FSB.org: https://www.fsb.org/work-of-the-fsb/market-and-institutional-resilience/post-2008-financial-crisis-reforms/ending-too-big-to-fail/global-systemically-important-financial-institutions-g-sifis/

Gabauer, D. (2020). Retrieved from Econometric-code: https://sites.google.com/view/davidgabauer/econometric-code

Gabauer, D. (2020). Volatility impulse response analysis for DCC-GARCH models: The role of volatility transmission mechanisms. *Journal of Forecasting, 39(5)*, 788-796.

Glod, A. (2018). European Sovereign and Private Debt Crisis What Should We Learn? . *Acta Universitatis Danbius Economica*, 88-100.

Gobbo, F. (2020, May 1). *KPMG.com*. Retrieved from Implications of Banks Risk Management: https://home.kpmg/xx/en/blogs/home/posts/2020/05/implications-on-risk-management.html

Goodell, J. (2020). COVID-19 and finance: Agendas for future research. *Finance Research Letters. July Volume 35*, https://doi.org/10.1016/J.FRL.2020.101512.

Gottesman, A., & Leibrock, M. (2017). *Understanding Systemic Risk in Global Financial Markets.* Hoboken, New Jersey: John Wiley & Sons.

Hanif, W., Mensi, W., & Vo, X. (2021). Impacts of COVID-19 outbreak on the spillovers between US and Chinese stock sectors. *Finance Research Letters: Volume 40*, https://doi.org/10.1016/J.FRL.2021.101922.

Huang, C., Wang, Y., Li, X., Ren, L., Zhao, J., Hu, Y., . . . Liu, M. (2020). *Clinical features of patients infected with 2019 novel coronavirus in Wuhan.* The Lancet, 395(10223), 497–506.

Jorda, Schularick, & Taylor. (2020). The effects of quasi-random monetary experiments. *Journal of Monetary Economics*, 22-40.

Kenton, W. (2020, December 18). *Investopedia.com*. Retrieved from Autoregressive Conditional Heteroskedasticity (ARCH): https://www.investopedia.com/terms/a/autoregressive-conditional-heteroskedasticity.asp

Kenton, W., & Potters, C. (2020, September 29). *Investopedia.com*. Retrieved from https://www.investopedia.com/terms/s/spillover-effect.asp

Laborda, R., & Olmo, J. (2021). Volatility spillover between economic sectors in financial crisis prediction: Evidence spanning the great financial crisis and Covid-19 pandemic. *Research in International Business and Finance*, https://doi.org/10.1016/j.ribaf.2021.101402.

Laborda, R., & Olmo, J. (2021). Volatility spillover between economic sectors in financial crisis prediction: Evidence spanning the great financial crisis and Covid-19 pandemic. *Research in International Business and Finance. Volume 57*, https://doi.org/10.1016/j.ribaf.2021.101402.

Lagoarde-Segot, T., & Leoni, P. (2013). Pandemics of the poor and banking stability. *Journal of Banking & Finance. Volume 37, Issue 11*, P. 4574-4583. https://doi.org/10.1016/j.jbankfin.2013.04.004.

Liu, X., Liu, Y., Hong, Y., & Wang, S. (2015). *Information Spillover Effect and Autoregressive Conditional Duration Models.* Routledge Taylor & Francis Group.

Ortmans, A., & Tripier, F. (2021). COVID-induced sovereign risk in the euro area: When did the ECB stop the spread? *European Economic Review: 137*, https://doi.org/10.1016/j.euroecorev.2021.103809.

Prabhakaran, S. (2019, July 2019). *Machinelearningplus.com*. Retrieved from Vector Autoregression (VAR) – Comprehensive Guide with Examples in Python: https://www.machinelearningplus.com/time-series/vector-autoregression-examples-python/

Reinhart, Carmen, Rogoff, & Kenneth. (2009). *This Time Is Different: Eight Centuries of Financial Folly.* Princeton, NJ: Princeton University Press.

Santis, R. D. (2013). The euro area sovereign debt crisis: Identifying flight-to-liquidity and the spillover mechanisms. *Journal of Empirical Finance Vol:26*, 150-170.

Saunders, K., Mark, N., Lewis, P., & Thornhill, A. (2019). *Research Methods for Business Students: Eight Edition.* Pearson.

Schildbach, J. (2020). *European banks suffer more than US peers in the corona crisis.* Deutsche Bank AG.

Shahzad, S., Naeem, M., Peng, Z., & Bouri, E. (2021). Asymmetric volatility spillover among Chinese sectors during COVID-19. *International Review of Financial Analysis. Volume 75, May*, https://doi.org/10.1016/j.irfa.2021.101754.

Shiller, R., Kon-Ya, F., & Tsutsui, Y. (1996). Why Did the Nikkei Crash? Expanding the Scope of Expectations Data Collection. *The Review of Economics and Statistics*, Vol.78, No.1 pp. 156-164.

Shu-Lien, C., & Yun-Huan, L. (2019). Returns spillovers between tourism ETFs. *North American Journal of Economics and Finance: 50*, https://doi.org/10.1016/j.najef.2019.04.020.

Skoufias, E. (2003). Economic Crises and Natural Disasters: Coping Strategies and Policy Implications. *World Development. Volume 31, Issue 7*, P. 1087-1102. https://doi.org/10.1016/S0305-750X(03)00069-X.

Statista Research Deparment. (2021). *Impact of COVID-19 on the global financial markets - Statistics & Facts, Feb 3, 2021*. Retrieved from https://www.statista.com/topics/6170/impact-of-covid-19-on-the-global-financial-markets/

Statista.com. (2021). *Statista.com*. Retrieved from Number of coronavirus (COVID-19) cases in Europe as of August 29, 2021, by country: https://www.statista.com/statistics/1104837/coronavirus-cases-europe-by-country/

Syllignakis, M., & Kouretas, G. (2011). Dynamic correlation analysis of financial conta- gion: evidence from the Central and Eastern European markets. *International Review of Economics & Finance: 20*, 717-732.

Tableau.com. (2020). Retrieved from https://www.tableau.com/learn/articles/time-series-analysis

Trichet, J. (2009). Ex-president of the European Central Bank. *Text of the Clare Distinguished Lecture in Economics and Public Policy.* Clare College, University of Cambridge.

Valdez, S., & Molyneux, P. (2015). *An Introduction to Global Financial Markets (8th edition).* London: Palgrave Macmillian.

Wang, H., & Li, S. (2021). Asymmetric volatility spillovers between crude oil and China's financial markets. *Energy Vol.233*, https://doi.org/10.1016/j.energy.2021.121168.

Washington Post. (2020). US intelligence reports from January and February warned about a likely pandemic. In *https://www.washingtonpost.com/national-security/us-intelligence-reports-from-january- and-february-warned-about-a-likely-pandemic/2020/03/20/299d8cda-6ad5-11ea-b5f1-a5a804158597_story.html.*

Welfens, P. (2008). Banking crisis and prudential supervision: a European perspective. *International Economics and Economic Policy*, 347-357.

Wijffelaars, M., & Loman, H. (2015). *The eurozone (debt) crisis – causes and crisis response*. Retrieved from Rabobank.com: https://economics.rabobank.com/publications/2015/december/the-eurozone-debt-crisis--causes-and-crisis-response/

Appendix B: Value at Risk

$$VaR = [R_p - (z)(\sigma)]\, V_p$$

R_p= Portfolio return
Z= Level of confidence
σ= Standard deviation
V_p= Portfolio weight

SOURCE: VERISTRAT.COM (2020)

Appendix C: Conditional Value at Risk

$$CVaR = \frac{1}{1-c}\int_{-1}^{VaR} xp(x)\,dx$$

Where:
p(x)dx = Probability density of return value "x"

c = Cut-off point on the distribution where VaR breakpoint is appointed

VaR = The appointed VaR level

Conditional Value at Risk. Source: Investopedia (2020)

Appendix D: Vector Autoregression

$$Y_t = \alpha + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \cdots + \beta_p Y_{t-p} + \epsilon_t$$

β = lag coefficients

α= a constant

Y= Time series

T = Time

ε= Error (white noise)

p= Predictors

Vector Autoregression. Source: Investopedia (2020)

Appendix E: DCC-GARCH (1,1) Part 1

1) $$\boldsymbol{y}_t = \boldsymbol{\mu}_t + \boldsymbol{\epsilon}_t \quad \boldsymbol{\epsilon}_t | \boldsymbol{F}_{t-1} \sim N(\boldsymbol{0}, \boldsymbol{H}_t),$$

2) $$\boldsymbol{\epsilon} = \boldsymbol{H}_t^{1/2} \boldsymbol{u}_t \quad \boldsymbol{u}_t \sim N(\boldsymbol{0}, \boldsymbol{I}),$$

3) $$\boldsymbol{H}_t = \boldsymbol{D}_t \boldsymbol{R}_t \boldsymbol{D}_t,$$

$\boldsymbol{F}_{t-1}$ = Available information up to $t-1$.

$\boldsymbol{y}_t, \boldsymbol{\mu}_t, \boldsymbol{\epsilon}_t, \boldsymbol{u}_t = N \times 1$- dimensional vectors describing analyzed time series, error term, conditional mean, and standardized error term

$\boldsymbol{R}_t, \boldsymbol{H}_t, \boldsymbol{D}_t = \text{diag}(h_{11t}^{1/2}, \ldots, h_{NNt}^{1/2}) = N \times N$ dimensional matrices, representing dynamic conditional correlations, time-varying conditional variances, and time-varying conditional variances.

Accordingly, a single shock and persistency parameter are expected

4) $$h_{ii,t} = \omega + \alpha \epsilon_{i,t-1}^2 + \beta h_{ii,t-1}.$$

The second stage presents the dynamic conditional correlations:

Source: Gabauer (2020)

Appendix E: DCC-GARCH (1,1) Part 2

$$\boldsymbol{R}_t = \operatorname{diag}\left(q_{iit}^{-1/2}, \ldots, q_{NNt}^{-1/2}\right) \boldsymbol{Q}_t \operatorname{diag}\left(q_{iit}^{-1/2}, \ldots, q_{NNt}^{-1/2}\right), \quad 5)$$

$$\boldsymbol{Q}_t = (1 - a - b)\bar{\boldsymbol{Q}} + a\boldsymbol{u}_{t-1}\boldsymbol{u}'_{t-1} + b\boldsymbol{Q}_{t-1}, \quad 6)$$

$\boldsymbol{Q}_t, \bar{\boldsymbol{Q}} = N \times N$ -dimensional positive-definitive matrices representing conditional and unconditional residuals' variance-covariance matrices.

a,(α), b (β) = nonnegative shock and persistency parameters fulfilling $a + b < 1$ ($\alpha + \beta \leq 1$) on condition if $a + b < 1$

$\boldsymbol{Q}_t$, $\boldsymbol{R}_t$ = Varying over time

$\boldsymbol{R}_t$ = Constant over time

The VIRF presents the shock impact in variable i on variable j's conditional volatilities which can be described as:

$$\boldsymbol{\Psi}^{\mathrm{g}} = \mathrm{VIRF}(J, \boldsymbol{\delta}_{j,t}, \boldsymbol{F}_{t-1}) = E(\boldsymbol{H}_{t+J} | \epsilon_{j,t} = \boldsymbol{\delta}_{j,t}, \boldsymbol{F}_{t-1}) - E(\boldsymbol{H}_{t+J} | \epsilon_{j,t} = \boldsymbol{0}, \boldsymbol{F}_{t-1}), \quad 7)$$

$\boldsymbol{\delta}_{j,t}$ = Selection vector with a one at the j position, or zero

where $\boldsymbol{\delta}_{j,t}$ is a selection vector with a one at the jth position and zero otherwise.

Source: Gabauer (2020)

Appendix E: DCC-GARCH (1,1) Part 3

Using the DCC=GARCH model to forecast conditional variance-covariance lies at the core of VIRF and can be carried out in three steps. The first step is calculated by GARCH (1, 1) and will forecast conditional volatilities ($\boldsymbol{D}_{t+h}|\boldsymbol{F}_t$) by:

$$E(h_{ii,t+1}|\boldsymbol{F}_t) = \omega + \alpha\delta_{1,t}^2 + \beta h_{ii,t} h = 1, \quad 8)$$

$$E(h_{ii,t+h}|\boldsymbol{F}_t) = \sum_{i=0}^{h-1} \omega(\alpha+\beta)^i + (\alpha+\beta)^{h-1} E(h_{ii,t+h-1}|\boldsymbol{F}_t) h > 1, \quad 9)$$

While in the second step $E(\boldsymbol{Q}_{t+h}|\boldsymbol{F}_t)$ is forecasted according to

$$E(\boldsymbol{Q}_{t+1}|\boldsymbol{F}_t) = (1-a-b)\bar{\boldsymbol{Q}} + a\boldsymbol{u}_t\boldsymbol{u}_t' + b\boldsymbol{Q}_t h = 1, \quad 10)$$

$$E(\boldsymbol{Q}_{t+h}|\boldsymbol{F}_t) = (1-a-b)\bar{\boldsymbol{Q}} + aE(\boldsymbol{u}_{t+h-1}\boldsymbol{u}_{t+h-1}'|\boldsymbol{F}_t) + bE(\boldsymbol{Q}_{t+h-1}|\boldsymbol{F}_t) h > 1, \quad 11)$$

Source: Gabauer (2020)

Appendix E: DCC-GARCH (1,1) Part 4

$E(\boldsymbol{u}_{t+h-1}\boldsymbol{u}'_{t+h-1}|\boldsymbol{F}_t) \approx E(\boldsymbol{Q}_{t+h-1}|\boldsymbol{F}_t)$ Supports in forecasting dynamic conditional correlations and conditional variance-covariance:

$$E(\boldsymbol{R}_{t+h}|\boldsymbol{F}_t) \approx \text{diag}\left[E(q_{iit+h}^{-1/2}|\boldsymbol{F}_t), \ldots, E(q_{NNt+h}^{-1/2}|\boldsymbol{F}_t)\right]$$
$$E(\boldsymbol{Q}_{t+h})\text{diag}\left[E\left(q_{iit+h}^{-1/2}|\boldsymbol{F}_t\right), \ldots, E\left(q_{NNt+h}^{-1/2}|\boldsymbol{F}_t\right),\right.$$

12)

$$E(\boldsymbol{H}_{t+h}|\boldsymbol{F}_t) \approx E(\boldsymbol{D}_{t+h}|\boldsymbol{F}_t)E(\boldsymbol{R}_{t+h}|\boldsymbol{F}_t)E(\boldsymbol{D}_{t+h}|\boldsymbol{F}_t).$$

13)

Established on the VIRF, the generalized forecast error variance decomposition is calculated. This can be explained as the variance share that a certain variable tells on others. The next steps to make variables explain 100 percent of variables $\boldsymbol{i}$'s forecast error variance is explained as follows:

$$\tilde{\phi}_{ij,t}^{g}(J) = \frac{\sum_{t=1}^{J-1}\Psi_{ij,t}^{2,g}}{\sum_{j=1}^{N}\sum_{t=1}^{J-1}\Psi_{ij,t}^{2,g}},$$

14)

$\sum_{j=1}^{N}\tilde{\phi}_{ij,t}^{g}(J) = 1$ and $\sum_{i,j=1}^{N}\tilde{\phi}_{ij,t}^{g}(J) = N$. The numerator stands for cumulative effect of the $\boldsymbol{i}$ shock, whereas the denominator presents aggregate cumulative effects of all shocks. The total connectedness index (TCI) based on GFEVD can be build up by:

Source: Gabauer (2020)

Appendix E: DCC-GARCH (1,1) Part 5

$$C_t^g(J) = \frac{\sum_{i,j=1, i\neq j}^{N} \tilde{\phi}_{ij,t}^g(J)}{N}. \quad 15)$$

Consequently, the variable i transmits to j. These are called 'total directional connectedness TO others' and are calculated as follows:

$$C_{i\rightarrow j,t}^g(J) = \frac{\sum_{j=1, i\neq j}^{N} \tilde{\phi}_{ji,t}^g(J)}{\sum_{j=1}^{N} \tilde{\phi}_{ji,t}^g(J)}. \quad 16)$$

The following step, variable i receives from j and is called 'total directional connectedness FROM others'. The calculation is as follows:

$$C_{i\leftarrow j,t}^g(J) = \frac{\sum_{j=1, i\neq j}^{N} \tilde{\phi}_{ij,t}^g(J)}{\sum_{i=1}^{N} \tilde{\phi}_{ij,t}^g(J)}. \quad 17)$$

By subtracting the previous mentioned measures from each other, the 'net total directional connectedness' can be explained as the substantial variable i has on the network:

$$C_{i,t}^g = C_{i\rightarrow j,t}^g(J) - C_{i\leftarrow j,t}^g(J). \quad 18)$$

In case net total directional connectedness from variable i is positive, it indicates that variable i is a net transmitter of shocks, or variable i is influenced by the network

Lastly, net pairwise directional connectedness (NPDC) from variable i and variable j is calculated as follows:

$$\mathrm{NPDC}_{ij}(J) = \tilde{\phi}_{ji,t}^g(J) - \tilde{\phi}_{ij,t}^g(J), \quad 19)$$

Here, a positive NPDC_{ij} explains that i dominates variable j

Source: Gabauer (2020)

Appendix F: Historical crises

Year	Event	County/Region	Broad Category
1636	Dutch Tulip Crisis	Europe	Asset Price Bubbles
1720	South Street Sea Bubble	Europe	Speculative Mania
1763	End of Seven Years War	Amsterdam	Asset Price Bubbles
1825	Crisis of 1825–1826	Europe/Latin America	Sovereign Default
1837	Crisis of 1836–1839	America/England	Price of Cotton
1857	Hamburg Crisis of 1857	Sweden/Hamburg	Expansion of Credit
1873	Panic of 1873	U.S., Austria, Germany	Global Contagion
1907	Panic of 1907	Global	Banking Crisis
1929	Great Depression	U.S./Europe	Banking Crisis
1977	"Big Five" Crisis	Spain	Real Estate Bubble/ Banking Crisis
1980s	Debt Crisis of the 1980s	U.S.	Sovereign Default, Currency Crash
1987	"Big Five" Crisis	Norway	Real Estate Bubble/ Banking Crisis
1990s	"Big Five" Crisis	Finland, Sweden, Japan	Real Estate Bubble/ Banking Crisis
1990s	Junk Bond Market Crash	U.S.	Asset Price Bubbles
1994	Mexican Debt Crisis	Mexico	Currency/Banking Crisis
1997	Asian Financial Crisis	Asia	Currency Crash
1998	Long-Term Capital Mgt.	U.S.	Credit
1990s	Latin American Debt Crisis	Latin America	Sovereign Default
2000	Dot-Com Tech Bubble	U.S.	Asset Bubble
2008	Credit Crisis	U.S./Europe	Asset Bubble

Timeline of historical crises. Source: Gottesman, A., & Leibrock, M. (2017). Understanding Systemic Risk in Global Financial Markets